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Research paper

Controllability and observability for linear quaternion-valued impulsive differential equations ☆

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ABSTRACT

In this paper, the controllability and observability for linear quaternion-valued impulsive differential equations (QIDEs) are investigated. We mainly establish some sufficient and necessary conditions for state controllability and state observability of linear QIDEs. By virtue of the isomorphism between quaternion vector space and complex variables space as well as the adjoint matrix of quaternion matrix, all the theoretical results in the sense of complex-valued and quaternion-valued are equivalent to each other. Finally, the validity of theoretical results acquired is indicated by two instances shown.

1. Introduction

Over the last few years, quaternion-valued differential equations (QDEs) have become a new research topic which are recognized as a special class of differential equations due to the non-exchangeability of quaternion multiplication. Although the research of QDEs is more complicated than ordinary differential equations, QDEs not only have been applied to Frenet frame of differential geometry [1–3], fluid mechanics [4–8], kinematic modeling [9] and quantum mechanics [10–13], but also have potential and advantages to characterize some processes related to multidimensional data, such as, attitude dynamics [14], spatial rigid body dynamics [15] and color image compression [16], etc. In consideration of theoretical and practical applications, QDEs have been developed and aroused more attention and research enthusiasms of a number of domestic and foreign researchers and experts. There are several published papers trying to study QDEs. Kou et al. [17,18] gave the basic theory and fundamental results for linear QDEs. Huang and He [19] established the second Lyapunov method over quaternion field which was used to consider the stability of autonomous and periodic nonautonomous systems. By virtue of the related space theorem and fixed point theorem, Yang and Ren [20] considered the initial value problem of quaternion fuzzy fractional differential equations. Chen et al. [21–24] dealt with the stability of

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1 The contributions of all authors (L.S., M.F. and J.W.) are equal. All the main results were developed together.

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linear QDEs. Fu et al. [25] derived the solutions of homogeneous and nonhomogeneous linear QDEs with delay via delayed quaternion matrix exponential and the method of variation of constants. Kou et al. [26] solved the linear QDEs which have multiple eigenvalues. Cai and Kou [27] proposed a special method adjoint matrix technique to solve the QDEs with two-sided coefficients and [28] solved linear QDEs by means of the method Laplace transform. Xia et al. [29] presented an algorithm to find a solution of linear nonhomogeneous QDEs, and [30] established the theory and applications of QDEs. Walcher and Zhang [31] further investigated the fundamental properties (stationary points and periodic orbits) of autonomous polynomial differential equations over the quaternions. Zhang [32] considered the global structure of the quaternion Bernoulli equation $\dot{q} = aq + bq^n$ for $q \in \mathbb{H}$.

Many processes are distinguished by abrupt changes or interference at certain moments, which is described as the impulsive phenomenon. Impulsive phenomenon is a noticeable and inevitable phenomenon in nature and human social activities, such as sudden social events on financial stock market, sudden change of external environment on the evolution of biological population and abnormal atmospheric circulation on the movement of flying objects, etc. Impulsive differential equations are suitable to depict abrupt changes or interference in practical problems, can fully consider the impact of sudden changes or interference on the system state, and accurately reflect the change rules and characteristics of many processes.

To the best of my knowledge, QDEs are usually used to depict the attitude dynamics of the moving object in the air, which can not only avoid the situation "Gimbal Lock" of the attitude dynamics equations expressed by Euler angle method at large angles, but also is more efficient. However, the moving object in the air may be interfered by the abnormal airflow, which will cause the state of the moving object to change abruptly. The classical QDEs are not suitable to depict this kind of phenomenon, and QIDEs are more appropriate to character it. Therefore, some scholars researched the QIDEs. Suo et al. [33] deduced the representations of the solutions for QIDEs in the sense of the complex-valued and quaternion-valued, respectively, and [34] considered periodic solutions of QIDEs. By means of utilizing Bielecki norm and fixed point theorem, Zhu and Sun [35] considered the existence and uniqueness of solutions for nonlinear QIDEs. Kashkynbayev and Mustafa [36] proved the analogue of the Floquet theorem for linear periodic QIDEs. Moreover, they studied the existence of bounded solutions and periodic solutions for linear nonhomogeneous QIDEs. Udhayakumar et al. [37,38] studied stability of quaternion-valued neural networks with impulses.

At the same time, it is common knowledge that the study on controllability and observability of impulsive differential equations or QDEs is necessary, which have been used in some branches of science and engineering, and a number of mathematical work involving them has been published by utilizing different approaches. Lupulescu and Younus [39] established sufficient and necessary criteria for the controllability and observability of linear time-varying and time-invariant impulsive control systems. Zhao and Sun [40] investigated the controllability and observability of linear time-varying impulsive systems, [41] defined the controllability and observability of time-varying impulsive systems in complex fields and gave several sufficient and necessary criteria for state controllability and observability. Xie and Wang [42] studied the controllability and observability of linear piecewise constant impulsive systems. Guan et al. [43] studied the controllability and observability for a class of time-varying impulsive control systems. Several sufficient and necessary conditions for state controllability and state observability of such systems were established and the corresponding criteria for time-invariant impulsive control systems were also obtained. Jiang et al. [44] defined an extension of the controllability and observability for linear QDEs and some criteria for controllability and observability were created. Chen et al. [45] adopted a special method to deal with the controllability and observability of linear QDEs from the point of complex-valued systems.

As far as we are aware, there is little work on state controllability and state observability for linear QIDEs and we would like to focus on them. The main contribution of this paper is to derive several sufficient and necessary conditions for state controllability and state observability of linear QIDEs. Most notably, all results obtained are presented in the sense of the complex-valued and quaternion-valued, which are equivalent to each other, respectively.

The rest of this paper is arranged as follows. In Section 2, some preliminaries are introduced. In Section 3, several sufficient and necessary criteria for state controllability of linear QIDEs are established. In Section 4, several sufficient and necessary criteria for state observability of linear QIDEs are studied. In the final section, two instances are shown to indicate the validity of our theoretical results.

2. Preliminaries

$\mathbb{R}$, $\mathbb{C}$ and $\mathbb{H}$ represent real numbers, complex numbers and quaternions, respectively. $\mathbb{C}^{n\times 1}$ and $\mathbb{H}^{n\times 1}$ represent $n \times 1$ complex vectors and $n \times 1$ quaternion vectors, respectively. $\mathbb{C}^{n\times n}$ and $\mathbb{H}^{n\times n}$ represent $n \times n$ complex matrices and $n \times n$ quaternion matrices, respectively.

A quaternion can be written as $q = q_0 + q_1\boldsymbol{i} + q_2\boldsymbol{j} + q_3\boldsymbol{k}$, where $q_0, q_1, q_2, q_3 \in \mathbb{R}$ and $\boldsymbol{i}, \boldsymbol{j}, \boldsymbol{k}$ satisfy the following multiplication table:

$$\boldsymbol{i}^2 = \boldsymbol{j}^2 = \boldsymbol{k}^2 = \boldsymbol{ijk} = -1, \boldsymbol{ij} = -\boldsymbol{ji} = \boldsymbol{k}, \boldsymbol{jk} = -\boldsymbol{kj} = \boldsymbol{i}, \boldsymbol{ki} = -\boldsymbol{ik} = \boldsymbol{j}.$$

The conjugate, the norm and the inverse of a quaternion q are $\overline{q} = q_0 - q_1\boldsymbol{i} - q_2\boldsymbol{j} - q_3\boldsymbol{k}$, $|q| = \sqrt{q\overline{q}} = \sqrt{q_0^2 + q_1^2 + q_2^2 + q_3^2}$ and $q^{-1} = \frac{\overline{q}}{|q|^2}$, respectively. For any two quaternions p and q, $\overline{pq} = \overline{q}\,\overline{p}$ is always satisfied, but it is worth noting that $pq \neq qp$.

$q = q_0 + q_1\boldsymbol{i} + q_2\boldsymbol{j} + q_3\boldsymbol{k} \in \mathbb{H}$ can be expressed as $q = c_1 + c_2\boldsymbol{j}$, where $c_1 = q_0 + q_1\boldsymbol{i}$, $c_2 = q_2 + q_3\boldsymbol{i} \in \mathbb{C}$. $A = (a_{ij}) \in \mathbb{H}^{n\times n}$ can be represented as $A = A_1 + A_2\boldsymbol{j}$, where $A_1, A_2 \in \mathbb{C}^{n\times n}$. Let

$$\zeta(A) = \begin{pmatrix} A_1 & A_2 \\ -\overline{A_2} & \overline{A_1} \end{pmatrix}$$

represent the $2n \times 2n$ adjoint matrix of A. $\upsilon \in \mathbb{H}^{n\times 1}$ can be expressed as $\upsilon = \upsilon_1 + \upsilon_2\boldsymbol{j}$, where $\upsilon_1, \upsilon_2 \in \mathbb{C}^{n\times 1}$. Let

$$\varphi(\upsilon) = \begin{pmatrix} \upsilon_1 \\ -\overline{\upsilon_2} \end{pmatrix},$$

where $\varphi : \upsilon \to \varphi(\upsilon)$ is an isomorphism mapping between $\mathbb{H}^{n\times 1}$ and $\mathbb{C}^{2n\times 1}$.

Let $C(J, \mathbb{H}^{n\times 1})$ denote the continuous quaternion-valued functions space from interval J into $\mathbb{H}^{n\times 1}$ equipped with the norm

$$\|x(t)\|_{\mathbb{H}^{n\times 1}} = \sqrt{\langle x(t), x(t)\rangle} = \sqrt{\sum_{i=1}^{n} \overline{x_i(t)}x_i(t)} = \sqrt{\sum_{i=1}^{n} |x_i(t)|^2}$$

and $PC(J, \mathbb{H}^{n\times 1}) = \{x : J \to \mathbb{H}^{n\times 1} : x \in C((t_k, t_{k+1}, \mathbb{H}^{n\times 1})$ and $x(t_k^-) = x(t_k), x(t_k^+)$ exist for every $k \in \mathbb{N})\}$ denote the piecewise continuous quaternion-valued functions space from interval J into $\mathbb{H}^{n\times 1}$ endowed with the norm

$$\|x\|_{\mathbb{PH}} = \sup_{t\in J} \|x(t)\|_{\mathbb{H}^{n\times 1}}.$$

3. Controllability

We discuss state controllability of the following linear QIDEs:

$$\begin{cases} \dot{x}(t) = Ax(t) + Cu(t),\ t \in J' := J\backslash\{t_k\},\ J := [t_0, \infty),\ k = 0, 1, 2, \ldots, \\ \Delta x(t_k) = Bx(t_k^-),\ k = 1, 2, \ldots, \\ y(t) = Dx(t), \end{cases} \tag{1}$$

where t_k satisfy $t_0 < t_1 < t_2 < \cdots < t_m < t_{m+1} < \cdots$, $x(\cdot) \in PC(J, \mathbb{H}^{n\times 1})$ is the state vector, $u(\cdot) \in PC(J, \mathbb{H}^{m\times 1})$ is the control input and $y(\cdot) \in PC(J, \mathbb{H}^{p\times 1})$ is the control output. $A, B \in \mathbb{H}^{n\times n}$, $C \in \mathbb{H}^{n\times m}$, $D \in \mathbb{H}^{p\times n}$. $x(t_k^+) = \lim_{\varepsilon\to 0^+} x(t_k + \varepsilon)$ and $x(t_k^-) = \lim_{\varepsilon\to 0^-} x(t_k + \varepsilon)$ represent the right and left limits of $x(t)$ at $t = t_k$.

Remark 3.1. Similar to the computation in [33], the solution $x(\cdot, x_0)$ of (1) with initial value condition $x(t_0) = x_0$ can be expressed by

$$\begin{aligned} x(t, x_0) = \varphi^{-1}\Bigg\{ & e^{\zeta(A)(t-t_k)} \prod_{j=k}^{1} \big[\zeta((E+B))e^{\zeta(A)(t_j - t_{j-1})}\big]\varphi(x_0) \\ & + e^{\zeta(A)(t-t_k)} \sum_{i=1}^{k} \prod_{j=k}^{i} \big[\zeta((E+B))e^{\zeta(A)(t_j - t_{j-1})}\big] \int_{t_{i-1}}^{t_i} e^{\zeta(A)(t_{i-1}-s)}\zeta(C)\varphi(u(s))ds \\ & + \int_{t_k}^{t} e^{\zeta(A)(t-s)}\zeta(C)\varphi(u(s))ds \Bigg\},\ t_k < t \le t_{k+1}, \end{aligned} \tag{2}$$

or

$$\begin{aligned} x(t, x_0) = {} & e^{A(t-t_k)} \prod_{j=k}^{1} \big[(E+B)e^{A(t_j - t_{j-1})}\big] x_0 \\ & + e^{A(t-t_k)} \sum_{i=1}^{k} \prod_{j=k}^{i} \big[(E+B)e^{A(t_j - t_{j-1})}\big] \int_{t_{i-1}}^{t_i} e^{A(t_{i-1}-s)}Cu(s)ds \\ & + \int_{t_k}^{t} e^{A(t-s)}Cu(s)ds,\ t_k < t \le t_{k+1}, \end{aligned} \tag{3}$$

where E is the identity matrix.

Definition 3.2 (*see [43, Definition 3.1]*)**.** The linear QIDEs (1) are called state controllable on $[t_0, t_l](t_l > t_0)$(or simply controllable if no confusion arises) if given any initial state $x(t_0) = x_0 \in \mathbb{H}^{n\times 1}$ there exists a piecewise continuous input signal $u(t) : [t_0, t_l] \to \mathbb{H}^{m\times 1}$ such that the corresponding solution of (1) satisfies $x(t_l) = 0$.

Lemma 3.3. *For $\forall\, A \in \mathbb{H}^{n\times n}$, then $\exists\, \beta_0(t), \beta_1(t), \ldots, \beta_{2n-1}(t)$ s.t.*

$$e^{At} = \sum_{j=0}^{2n-1} \beta_j(t) A^j, \tag{4}$$

where $\beta_0(t), \beta_1(t), \ldots, \beta_{2n-1}(t)$ are scalar functions.

Proof. According to (10) of [41], we know that for $\forall\, P \in \mathbb{C}^{n\times n}$, $\exists\, \beta_0(t), \beta_1(t), \cdots, \beta_{n-1}(t)$ s.t.

$$e^{Pt} = \sum_{j=0}^{n-1} \beta_j(t) P^j. \tag{5}$$

According to Theorem 4.2 of [46], we know that $\zeta(AB) = \zeta(A)\zeta(B)$ and $\zeta(A+B) = \zeta(A) + \zeta(B)$ hold. Therefore, we can derive that

$$\begin{aligned}(\zeta(A))^j &= \overbrace{\zeta(A)\zeta(A)\cdots\zeta(A)}^{j} \\ &= \zeta(\overbrace{A\cdots A}^{j}) \\ &= \zeta(A^j)\end{aligned} \tag{6}$$

and

$$\begin{aligned}\zeta(e^{At}) &= \zeta\left(\sum_{n=0}^{\infty} \frac{A^n t^n}{n!}\right) \\ &= \zeta\left(E + \frac{At}{1!} + \frac{A^2t^2}{2!} + \cdots\right) \\ &= \zeta\,(E) + \zeta\left(\frac{At}{1!}\right) + \zeta\left(\frac{A^2t^2}{2!}\right) + \cdots \\ &= \zeta(E) + \frac{\zeta(At)}{1!} + \frac{\zeta(A^2t^2)}{2!} + \cdots \\ &= \zeta(E) + \frac{\zeta(A)t}{1!} + \frac{\zeta(A^2)t^2}{2!} + \cdots \\ &= \zeta(E) + \frac{\zeta(A)t}{1!} + \frac{(\zeta(A))^2t^2}{2!} + \cdots \\ &= \sum_{n=0}^{\infty} \frac{(\zeta(A))^n t^n}{n!} \\ &= e^{\zeta(A)t}\end{aligned} \tag{7}$$

hold.

Hence, for $\forall\, A \in \mathbb{H}^{n\times n}$, by means of (5), (6) and (7), we can obtain that

$$\begin{aligned}e^{At} &= \zeta^{-1}\left(\zeta(e^{At})\right) \\ &= \zeta^{-1}\left(e^{\zeta(A)t}\right) \\ &= \zeta^{-1}\left(\sum_{j=0}^{2n-1} \beta_j(t)\,(\zeta(A))^j\right) \\ &= \sum_{j=0}^{2n-1} \beta_j(t) A^j. \quad \square\end{aligned}$$

Here, define $n \times n$ and $2n \times 2n$ matrices as below:

$$\begin{aligned}\overline{V}_{i-1} &:= \overline{V}(t_{i-1}, t_i) \\ &= \int_{t_{i-1}}^{t_i} e^{\zeta(A)(t_{i-1}-s)} \zeta(C)\zeta(C^*) e^{\zeta(A^*)(t_{i-1}-s)} ds, \\ &\quad i = 1, \ldots, k,\end{aligned}$$

$$
\begin{aligned}
\overline{V}_k &:= \overline{V}(t_k, t_l) \\
&= \int_{t_k}^{t_l} e^{\zeta(A)(t_k-s)}\zeta(C)\zeta(C^*)e^{\zeta(A^*)(t_k-s)}ds, \\
\overline{U}_0 &:= \overline{U}(t_0, t_1) \\
&= \int_{t_0}^{t_1} e^{\zeta(A)(t_0-s)}\zeta(C)\zeta(C^*)e^{\zeta(A^*)(t_0-s)}ds, \\
\overline{U}_{i-1} &:= \overline{U}(t_{i-1}, t_i) \\
&= \int_{t_{i-1}}^{t_i} \prod_{j=1}^{i-1}\big[e^{-\zeta(A)(t_j-t_{j-1})}\zeta((E+B)^{-1})\big]e^{\zeta(A)(t_{i-1}-s)}\zeta(C) \\
&\quad \times \zeta(C^*)e^{\zeta(A^*)(t_{i-1}-s)}\prod_{j=1}^{i-1}\big[\zeta((E+B^*)^{-1})e^{-\zeta(A^*)(t_j-t_{j-1})}\big]ds, \\
&\quad i = 2, \ldots, k, \\
\overline{U}_k &:= \overline{U}(t_k, t_l) \\
&= \int_{t_k}^{t_l} \prod_{j=1}^{k}\big[e^{-\zeta(A)(t_j-t_{j-1})}\zeta((E+B)^{-1})\big]e^{\zeta(A)(t_k-s)}\zeta(C) \\
&\quad \times \zeta(C^*)e^{\zeta(A^*)(t_k-s)}\prod_{j=1}^{k}\big[\zeta((E+B^*)^{-1})e^{-\zeta(A^*)(t_j-t_{j-1})}\big]ds, \\
V_{i-1} &:= V(t_{i-1}, t_i) \\
&= \int_{t_{i-1}}^{t_i} e^{A(t_{i-1}-s)}CC^*e^{A^*(t_{i-1}-s)}ds, \\
&\quad i = 1, \ldots, k, \\
V_k &:= V(t_k, t_l) \\
&= \int_{t_k}^{t_l} e^{A(t_k-s)}CC^*e^{A^*(t_k-s)}ds, \\
U_0 &:= U(t_0, t_1) \\
&= \int_{t_0}^{t_1} e^{A(t_0-s)}CC^*e^{A^*(t_0-s)}ds, \\
U_{i-1} &:= U(t_{i-1}, t_i) \\
&= \int_{t_{i-1}}^{t_i} \prod_{j=1}^{i-1}\big[e^{-A(t_j-t_{j-1})}(E+B)^{-1}\big]e^{A(t_{i-1}-s)}C \\
&\quad \times C^*e^{A^*(t_{i-1}-s)}\prod_{j=1}^{i-1}\big[(E+B^*)^{-1}e^{-A^*(t_j-t_{j-1})}\big]ds, \\
&\quad i = 2, \ldots, k, \\
U_k &:= U(t_k, t_l) \\
&= \int_{t_k}^{t_l} \prod_{j=1}^{k}\big[e^{-A(t_j-t_{j-1})}(E+B)^{-1}\big]e^{A(t_k-s)}C \\
&\quad \times C^*e^{A^*(t_k-s)}\prod_{j=1}^{k}\big[(E+B^*)^{-1}e^{-A^*(t_j-t_{j-1})}\big]ds,
\end{aligned} \tag{8}
$$

where A^*, B^* and C^* denote the conjugate transpose of A, B and C respectively.

Theorem 3.4. *If there is at least a matrix $\overline{V}_f (f \in \{0, 1, \ldots, k\})$ such that $\mathrm{rank}(\overline{V}_f) = 2n$, then the linear QIDEs (1) are state controllable on $[t_0, t_l](t_l \in (t_k, t_{k+1}])$.*

Proof. See Appendix A for the proof of Theorem 3.4. □

Theorem 3.5. *If there is at least a matrix $V_f (f \in \{0, 1, \ldots, k\})$ such that $rank(V_f) = n$, then the linear QIDEs (1) are state controllable on $[t_0, t_l](t_l \in (t_k, t_{k+1}])$.*

Proof. See Appendix B for the proof of Theorem 3.5. □

Theorem 3.6. *Suppose that $\zeta((E+B))$ is invertible. If the linear QIDEs (1) are state controllable on $[t_0, t_l](t_l \in (t_k, t_{k+1}])$, then*

$$rank(\overline{U}_0, \overline{U}_1, \ldots, \overline{U}_k) = 2n.$$

Proof. See Appendix C for the proof of Theorem 3.6. □

Theorem 3.7. *Suppose that $(E + B)$ is invertible. If the linear QIDEs (1) are state controllable on $[t_0, t_l](t_l \in (t_k, t_{k+1}])$, then*

$$rank(U_0, U_1, \ldots, U_k) = n.$$

Proof. See Appendix D for the proof of Theorem 3.7. □

Remark 3.8. Obviously, the linear QIDEs (1) without impulses become the following QDEs

$$\begin{cases} \dot{x}(t) = Ax(t) + Cu(t), \ t \in J, \ J := [t_0, \infty), \\ y(t) = Dx(t), \end{cases} \tag{9}$$

Correspondingly, (2) and (3) become

$$x(t, x_0) = \varphi^{-1}\left\{ e^{\zeta(A)(t-t_0)}\varphi(x_0) + \int_{t_0}^{t} e^{\zeta(A)(t-s)}\zeta(C)\varphi(u(s))ds \right\},$$

or

$$x(t, x_0) = e^{A(t-t_0)}x_0 + \int_{t_0}^{t} e^{A(t-s)}Cu(s)ds.$$

Letting $B = 0$, the results obtained for the state controllability of the linear QIDEs (1) are also applicable to QDEs (9). Theorem 3.4, Theorem 3.5, Theorem 3.6 and Theorem 3.7 for (1) turn into Remark 3.9 for (9). Theorem 3.10, Theorem 3.11, Theorem 3.12 and Theorem 3.13 for (1) turn into Remark 3.14 for (9).

Remark 3.9. The linear QDEs (9) are state controllable on $[t_0, t_l]$ if and only if

$$rank(\overline{V}) = rank\left(\int_{t_0}^{t_l} e^{\zeta(A)(t_0-s)}\zeta(C)\zeta(C^*)e^{\zeta(A^*)(t_0-s)}ds \right) = 2n$$

or

$$rank(V) = rank\left(\int_{t_0}^{t_l} e^{A(t_0-s)}CC^*e^{A^*(t_0-s)}ds \right) = n.$$

Theorem 3.10. *If*

$$rank(\overline{Q}_c) = rank(\zeta(C), \zeta(A)\zeta(C), \ldots, \zeta(A^{2n-1})\zeta(C)) = 2n, \tag{10}$$

then the linear QIDEs (1) are state controllable on $[t_0, t_l](t_l \in (t_k, t_{k+1}])$.

Proof. See Appendix E for the proof of Theorem 3.10. □

Theorem 3.11. *If*

$$rank(Q_c) = rank(C, AC, \ldots, A^{2n-1}C) = n, \tag{11}$$

then the linear QIDEs (1) are state controllable on $[t_0, t_l](t_l \in (t_k, t_{k+1}])$.

Proof. See Appendix F for the proof of Theorem 3.11. □

Theorem 3.12. *Suppose that $\zeta(A)\zeta(B) = \zeta(B)\zeta(A)$ and $\zeta((E + B))$ is invertible. If the linear QIDEs (1) are state controllable on $[t_0, t_l](t_l \in (t_k, t_{k+1}])$, then*

$$rank(\overline{Z}) = rank(\zeta((E + B)^k)\overline{Q}_c, \zeta((E + B)^{k-1})\overline{Q}_c, \ldots, \overline{Q}_c) = 2n,$$

where $\overline{Q}_c = (\zeta(C), \zeta(A)\zeta(C), \ldots, \zeta(A^{2n-1})\zeta(C))$.

Proof. See Appendix G for the proof of Theorem 3.12. □

Theorem 3.13. *Suppose that $AB = BA$ and $(E + B)$ is invertible. If the linear QIDEs (1) are state controllable on $[t_0, t_l](t_l \in (t_k, t_{k+1}])$, then*

$$rank(Z) = rank((E + B)^k Q_c, (E + B)^{k-1} Q_c, \ldots, Q_c) = n,$$

where $Q_c = (C, AC, \ldots, A^{2n-1}C)$.

Proof. See Appendix H for the proof of Theorem 3.13. □

Remark 3.14. The linear QDEs (9) are state controllable on $[t_0, t_l]$ if and only if

$$rank(\overline{Q}) = rank(\zeta(C), \zeta(A)\zeta(C), \ldots, \zeta(A^{2n-1})\zeta(C)) = 2n$$

or

$$rank(Q) = rank(C, AC, \ldots, A^{2n-1}C) = n.$$

4. Observability

Definition 4.1 (*see [43, Definition 4.1]*)**.** The linear QIDEs (1) are called state observable on $[t_0, t_l](t_l > t_0)$(or simply observable if no confusion arises) if any initial state $x(t_0) = x_0 \in \mathbb{H}^{n\times 1}$ is uniquely determined by the corresponding system input $u(t)$ and system output $y(t)$ for $t \in [t_0, t_l]$.

Here, define $n \times n$ and $2n \times 2n$ matrices as below:

$$\begin{aligned}
\overline{N}(t_0, t_l) &= \sum_{i=1}^{k} \overline{N}(t_{i-1}, t_i) + \overline{N}(t_k, t_l),\\
\overline{N}(t_0, t_1) &= \int_{t_0}^{t_1} e^{\zeta(A^*)(s-t_0)}\zeta(D^*)\zeta(D)e^{\zeta(A)(s-t_0)}ds,\\
\overline{N}(t_{i-1}, t_i) &= \int_{t_{i-1}}^{t_i} \prod_{j=i-1}^{1} \left[e^{\zeta(A^*)(t_j-t_{j-1})}\zeta((E+B^*))\right]e^{\zeta(A^*)(s-t_{i-1})}\zeta(D^*)\\
&\quad \times \zeta(D)e^{\zeta(A)(s-t_{i-1})} \prod_{j=i-1}^{1} \left[\zeta((E+B))e^{\zeta(A)(t_j-t_{j-1})}\right]ds,\\
&\quad i = 2, \ldots, k,\\
\overline{N}(t_k, t_l) &= \int_{t_k}^{t_l} \prod_{j=k}^{1}\left[e^{\zeta(A^*)(t_j-t_{j-1})}\zeta((E+B^*))\right]e^{\zeta(A^*)(s-t_k)}\zeta(D^*)\\
&\quad \times \zeta(D)e^{\zeta(A)(s-t_k)} \prod_{j=k}^{1}\left[\zeta((E+B))e^{\zeta(A)(t_j-t_{j-1})}\right]ds,\\
N(t_0, t_l) &= \sum_{i=1}^{k} N(t_{i-1}, t_i) + N(t_k, t_l),\\
N(t_0, t_1) &= \int_{t_0}^{t_1} e^{A^*(s-t_0)}D^*De^{A(s-t_0)}ds,\\
N(t_{i-1}, t_i) &= \int_{t_{i-1}}^{t_i} \prod_{j=i-1}^{1} \left[e^{A^*(t_j-t_{j-1})}(E+B^*)\right]e^{A^*(s-t_{i-1})}D^*\\
&\quad \times De^{A(s-t_{i-1})} \prod_{j=i-1}^{1} \left[(E+B)e^{A(t_j-t_{j-1})}\right]ds,\\
&\quad i = 2, \ldots, k,\\
N(t_k, t_l) &= \int_{t_k}^{t_l} \prod_{j=k}^{1}\left[e^{A^*(t_j-t_{j-1})}(E+B^*)\right]e^{A^*(s-t_k)}D^*\\
&\quad \times De^{A(s-t_k)} \prod_{j=k}^{1}\left[(E+B)e^{A(t_j-t_{j-1})}\right]ds,
\end{aligned} \tag{12}$$

where A^*, B^* and D^* denote the conjugate transpose of A, B and D respectively.

Theorem 4.2. *The linear QIDEs (1) are state observable on $[t_0, t_l](t_l \in (t_k, t_{k+1}])$, if and only if the quaternion matrix $\overline{N}(t_0, t_l)$ defined in (12) is invertible.*

Proof. See Appendix I for the proof of Theorem 4.2. □

Theorem 4.3. *The linear QIDEs (1) are state observable on $[t_0, t_l](t_l \in (t_k, t_{k+1}])$, if and only if the quaternion matrix $N(t_0, t_l)$ defined in (12) is invertible.*

Proof. See Appendix J for the proof of Theorem 4.3. □

Letting $B = 0$, the results obtained for the state observability of the linear QIDEs (1) are also effective to QDEs (9). Theorem 4.2 and Theorem 4.3 for (1) become Remark 4.4 for (9). Theorem 4.5, Theorem 4.6, Theorem 4.7 and Theorem 4.8 for (1) become Remark 4.9 for (9).

Remark 4.4. The linear QDEs (9) are state observable on $[t_0, t_l]$ if and only if

$$rank(\overline{W}) = rank\left(\int_{t_0}^{t_l} e^{\zeta(A^*)(s-t_0)}\zeta(D^*)\zeta(D)e^{\zeta(A)(s-t_0)}ds\right) = 2n$$

or

$$rank(W) = rank\left(\int_{t_0}^{t_l} e^{A^*(s-t_0)}D^*De^{A(s-t_0)}ds\right) = n.$$

Theorem 4.5. *If*

$$rank(\overline{Q}_o) = rank\begin{pmatrix} \zeta(D) \\ \zeta(D)\zeta(A) \\ \vdots \\ \zeta(D)\zeta(A^{2n-1}) \end{pmatrix} = 2n, \tag{13}$$

then the linear QIDEs (1) are state observable on $[t_0, t_l](t_l \in (t_k, t_{k+1}])$.

Proof. See Appendix K for the proof of Theorem 4.5. □

Theorem 4.6. *If*

$$rank(Q_o) = rank\begin{pmatrix} D \\ DA \\ \vdots \\ DA^{2n-1} \end{pmatrix} = n, \tag{14}$$

then the linear QIDEs (1) are state observable on $[t_0, t_l](t_l \in (t_k, t_{k+1}])$.

Proof. See Appendix L for the proof of Theorem 4.6. □

Theorem 4.7. *Suppose that $\zeta(A)\zeta(B) = \zeta(B)\zeta(A)$. If the linear QIDEs (1) are state observable on $[t_0, t_l](t_l \in (t_k, t_{k+1}])$, then*

$$rank(\overline{S}) = rank\begin{pmatrix} \overline{Q}_o \\ \overline{Q}_o\zeta((E+B)) \\ \vdots \\ \overline{Q}_o\zeta((E+B)^k) \end{pmatrix} = 2n,$$

where

$$\overline{Q}_o = \begin{pmatrix} \zeta(D) & \zeta(D)\zeta(A) & \cdots & \zeta(D)\zeta(A^{2n-1}) \end{pmatrix}^T.$$

Proof. See Appendix M for the proof of Theorem 4.7. □

Theorem 4.8. *Suppose that $AB = BA$. If the linear QIDEs (1) are state observable on $[t_0, t_l](t_l \in (t_k, t_{k+1}])$, then*

$$rank(S) = rank \begin{pmatrix} Q_o \\ Q_o(E+B) \\ \vdots \\ Q_o(E+B)^k \end{pmatrix} = n,$$

where

$$Q_o = \begin{pmatrix} D & DA & \cdots & DA^{2n-1} \end{pmatrix}^T.$$

Proof. See Appendix N for the proof of Theorem 4.8. □

Remark 4.9. The linear QDEs (9) are state observable on $[t_0, t_l]$ if and only if

$$rank(\overline{Q}) = rank \begin{pmatrix} \zeta(D) \\ \zeta(D)\zeta(A) \\ \vdots \\ \zeta(D)\zeta(A^{2n-1}) \end{pmatrix} = 2n,$$

or

$$rank(Q) = rank \begin{pmatrix} D \\ DA \\ \vdots \\ DA^{2n-1} \end{pmatrix} = n.$$

5. Examples

Example 5.1. Considering linear QIDEs:

$$\begin{cases} \dot{x}(t) = \begin{pmatrix} -1+\boldsymbol{j} & 0 \\ 0 & 2 \end{pmatrix} x(t) + \begin{pmatrix} 0 & \boldsymbol{i} \\ \boldsymbol{j} & 0 \end{pmatrix} u(t),\ t \in J' := [0, \frac{5}{2}]\backslash\{t_k\},\ k = 1, 2, \\ \Delta x(t_k) = \begin{pmatrix} 1 & 0 \\ 0 & -1+\boldsymbol{k} \end{pmatrix} x(t_k^-),\ k = 1, 2, \\ y(t) = \begin{pmatrix} 1 & 0 \\ 0 & \boldsymbol{k} \end{pmatrix} x(t), \\ x(0) = \begin{pmatrix} 1+\boldsymbol{i} \\ 1+\boldsymbol{i}+\boldsymbol{j}+\boldsymbol{k} \end{pmatrix}, \end{cases} \tag{15}$$

where $t_k = k$.

We have

$$A = \begin{pmatrix} -1+\boldsymbol{j} & 0 \\ 0 & 2 \end{pmatrix},\ B = \begin{pmatrix} 1 & 0 \\ 0 & -1+\boldsymbol{k} \end{pmatrix},\ C = \begin{pmatrix} 0 & \boldsymbol{i} \\ \boldsymbol{j} & 0 \end{pmatrix},\ D = \begin{pmatrix} 1 & 0 \\ 0 & \boldsymbol{k} \end{pmatrix},$$

$$x_0 = x(0) = \begin{pmatrix} 1+\boldsymbol{i} \\ 1+\boldsymbol{i}+\boldsymbol{j}+\boldsymbol{k} \end{pmatrix},$$

and then

$$\zeta(A) = \begin{pmatrix} -1 & 0 & 1 & 0 \\ 0 & 2 & 0 & 0 \\ -1 & 0 & -1 & 0 \\ 0 & 0 & 0 & 2 \end{pmatrix},\ \zeta(B) = \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & -1 & 0 & \boldsymbol{i} \\ 0 & 0 & 1 & 0 \\ 0 & \boldsymbol{i} & 0 & -1 \end{pmatrix},\ \zeta(C) = \begin{pmatrix} 0 & \boldsymbol{i} & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & -\boldsymbol{i} \\ -1 & 0 & 0 & 0 \end{pmatrix},$$

$$\zeta(D) = \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & \boldsymbol{i} \\ 0 & 0 & 1 & 0 \\ 0 & \boldsymbol{i} & 0 & 0 \end{pmatrix},\ \varphi(x_0) = \begin{pmatrix} 1+\boldsymbol{i} \\ 1+\boldsymbol{i} \\ 0 \\ -1+\boldsymbol{i} \end{pmatrix}.$$

From (8),

$$\begin{aligned}
\overline{V}_1 &= \overline{V}(t_1,t_2)\\
&= \int_1^2 e^{\zeta(A)(1-s)}\zeta(C)\zeta(C^*)e^{\zeta(A^*)(1-s)}ds\\
&= \int_1^2 \begin{pmatrix} e^{-(1-s)}\cos(1-s) & 0 & e^{-(1-s)}\sin(1-s) & 0\\ 0 & e^{2(1-s)} & 0 & 0\\ -e^{-(1-s)}\sin(1-s) & 0 & e^{-(1-s)}\cos(1-s) & 0\\ 0 & 0 & 0 & e^{2(1-s)} \end{pmatrix} \begin{pmatrix} 0 & \boldsymbol{i} & 0 & 0\\ 0 & 0 & 1 & 0\\ 0 & 0 & 0 & -\boldsymbol{i}\\ -1 & 0 & 0 & 0 \end{pmatrix}\\
&\quad \times \begin{pmatrix} 0 & 0 & 0 & -1\\ -\boldsymbol{i} & 0 & 0 & 0\\ 0 & 1 & 0 & 0\\ 0 & 0 & \boldsymbol{i} & 0 \end{pmatrix} \begin{pmatrix} e^{-(1-s)}\cos(1-s) & 0 & -e^{-(1-s)}\sin(1-s) & 0\\ 0 & e^{2(1-s)} & 0 & 0\\ e^{-(1-s)}\sin(1-s) & 0 & e^{-(1-s)}\cos(1-s) & 0\\ 0 & 0 & 0 & e^{2(1-s)} \end{pmatrix} ds\\
&= \begin{pmatrix} \frac{1}{2}(e^2-1) & 0 & 0 & 0\\ 0 & \frac{1}{4}(1-e^{-4}) & 0 & 0\\ 0 & 0 & \frac{1}{2}(e^2-1) & 0\\ 0 & 0 & 0 & \frac{1}{4}(1-e^{-4}) \end{pmatrix},\\
&\quad rank\ \overline{V}_1 = 4.
\end{aligned}$$

From (17),

$$\begin{aligned}
\varphi(u(t)) &= -\zeta(C^*)e^{\zeta(A^*)(1-t)}\overline{V}^{-1}(t_1,t_2)\zeta((E+B))e^{\zeta(A)}\varphi(x_0)\\
&= -\begin{pmatrix} 0 & 0 & 0 & -1\\ -\boldsymbol{i} & 0 & 0 & 0\\ 0 & 1 & 0 & 0\\ 0 & 0 & \boldsymbol{i} & 0 \end{pmatrix} \begin{pmatrix} e^{-(1-t)}\cos(1-t) & 0 & -e^{-(1-t)}\sin(1-t) & 0\\ 0 & e^{2(1-t)} & 0 & 0\\ e^{-(1-t)}\sin(1-t) & 0 & e^{-(1-t)}\cos(1-t) & 0\\ 0 & 0 & 0 & e^{2(1-t)} \end{pmatrix}\\
&\quad \times \begin{pmatrix} \frac{2}{e^2-1} & 0 & 0 & 0\\ 0 & \frac{4}{1-e^{-4}} & 0 & 0\\ 0 & 0 & \frac{2}{e^2-1} & 0\\ 0 & 0 & 0 & \frac{4}{1-e^{-4}} \end{pmatrix} \begin{pmatrix} 2 & 0 & 0 & 0\\ 0 & 0 & 0 & \boldsymbol{i}\\ 0 & 0 & 2 & 0\\ 0 & \boldsymbol{i} & 0 & 0 \end{pmatrix} \begin{pmatrix} e^{-1}\cos 1 & 0 & e^{-1}\sin 1 & 0\\ 0 & e^2 & 0 & 0\\ -e^{-1}\sin 1 & 0 & e^{-1}\cos 1 & 0\\ 0 & 0 & 0 & e^2 \end{pmatrix} \begin{pmatrix} 1+\boldsymbol{i}\\ 1+\boldsymbol{i}\\ 0\\ -1+\boldsymbol{i} \end{pmatrix}\\
&= \begin{pmatrix} \frac{4e^{-2t}}{e^{-4}(1-e^{-4})}(\boldsymbol{i}-1)\\ \frac{4e^t\cos t}{e^2(e^2-1)}(\boldsymbol{i}-1)\\ \frac{4e^{-2t}}{e^{-4}(1-e^{-4})}(\boldsymbol{i}+1)\\ \frac{4e^t\sin t}{e^2(e^2-1)}(\boldsymbol{i}-1) \end{pmatrix},
\end{aligned}$$

where $t\in(1,2]$.

When $t_l=\frac{5}{2}$,

$$\begin{aligned}
x(t_l,x_0) &= \varphi^{-1}\{e^{\frac{1}{2}\zeta(A)}\zeta((E+B))e^{\zeta(A)}\zeta((E+B))e^{\zeta(A)}\varphi(x_0)\\
&\quad + e^{\frac{1}{2}\zeta(A)}\zeta((E+B))e^{\zeta(A)}\int_1^2 e^{\zeta(A)(1-s)}\zeta(C)\varphi(u(s))ds\}\\
&= \varphi^{-1}\left\{ \begin{pmatrix} e^{-\frac{1}{2}}\cos\frac{1}{2} & 0 & e^{-\frac{1}{2}}\sin\frac{1}{2} & 0\\ 0 & e & 0 & 0\\ -e^{-\frac{1}{2}}\sin\frac{1}{2} & 0 & e^{-\frac{1}{2}}\cos\frac{1}{2} & 0\\ 0 & 0 & 0 & e \end{pmatrix} \begin{pmatrix} 2 & 0 & 0 & 0\\ 0 & 0 & 0 & \boldsymbol{i}\\ 0 & 0 & 2 & 0\\ 0 & \boldsymbol{i} & 0 & 0 \end{pmatrix} \begin{pmatrix} e^{-1}\cos 1 & 0 & e^{-1}\sin 1 & 0\\ 0 & e^2 & 0 & 0\\ -e^{-1}\sin 1 & 0 & e^{-1}\cos 1 & 0\\ 0 & 0 & 0 & e^2 \end{pmatrix} \right.
\end{aligned}$$

$$\times \begin{pmatrix} 2 & 0 & 0 & 0 \\ 0 & 0 & 0 & \boldsymbol{i} \\ 0 & 0 & 2 & 0 \\ 0 & \boldsymbol{i} & 0 & 0 \end{pmatrix} \begin{pmatrix} e^{-1}\cos 1 & 0 & e^{-1}\sin 1 & 0 \\ 0 & e^2 & 0 & 0 \\ -e^{-1}\sin 1 & 0 & e^{-1}\cos 1 & 0 \\ 0 & 0 & 0 & e^2 \end{pmatrix} \begin{pmatrix} 1+\boldsymbol{i} \\ 1+\boldsymbol{i} \\ 0 \\ -1+\boldsymbol{i} \end{pmatrix}$$

$$+ \begin{pmatrix} e^{-\frac{1}{2}}\cos\frac{1}{2} & 0 & e^{-\frac{1}{2}}\sin\frac{1}{2} & 0 \\ 0 & e & 0 & 0 \\ -e^{-\frac{1}{2}}\sin\frac{1}{2} & 0 & e^{-\frac{1}{2}}\cos\frac{1}{2} & 0 \\ 0 & 0 & 0 & e \end{pmatrix} \begin{pmatrix} 2 & 0 & 0 & 0 \\ 0 & 0 & 0 & \boldsymbol{i} \\ 0 & 0 & 2 & 0 \\ 0 & \boldsymbol{i} & 0 & 0 \end{pmatrix} \begin{pmatrix} e^{-1}\cos 1 & 0 & e^{-1}\sin 1 & 0 \\ 0 & e^2 & 0 & 0 \\ -e^{-1}\sin 1 & 0 & e^{-1}\cos 1 & 0 \\ 0 & 0 & 0 & e^2 \end{pmatrix}$$

$$\times \int_1^2 \begin{pmatrix} e^{-(1-s)}\cos(1-s) & 0 & e^{-(1-s)}\sin(1-s) & 0 \\ 0 & e^{2(1-s)} & 0 & 0 \\ -e^{-(1-s)}\sin(1-s) & 0 & e^{-(1-s)}\cos(1-s) & 0 \\ 0 & 0 & 0 & e^{2(1-s)} \end{pmatrix}$$

$$\times \begin{pmatrix} 0 & \boldsymbol{i} & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & -\boldsymbol{i} \\ -1 & 0 & 0 & 0 \end{pmatrix} \begin{pmatrix} \frac{4e^{-2s}}{e^{-4}(1-e^{-4})}(\boldsymbol{i}-1) \\ \frac{4e^{s}\cos s}{e^2(e^2-1)}(\boldsymbol{i}-1) \\ \frac{4e^{-2s}}{e^{-4}(1-e^{-4})}(\boldsymbol{i}+1) \\ \frac{4e^{s}\sin s}{e^2(e^2-1)}(\boldsymbol{i}-1) \end{pmatrix} ds \Bigg\}$$

$$= \begin{pmatrix} 0 \\ 0 \end{pmatrix}.$$

First, the initial value $x_0 = \begin{pmatrix} 1+\boldsymbol{i} \\ 1+\boldsymbol{i}+\boldsymbol{j}+\boldsymbol{k} \end{pmatrix}$ given, we intend to make the system (15) controlled at $t = \frac{5}{2}$. Next, we require to select the appropriate control $\varphi(u(t)) = \begin{pmatrix} \frac{4e^{-2t}}{e^{-4}(1-e^{-4})}(\boldsymbol{i}-1) \\ \frac{4e^{t}\cos t}{e^2(e^2-1)}(\boldsymbol{i}-1) \\ \frac{4e^{-2t}}{e^{-4}(1-e^{-4})}(\boldsymbol{i}+1) \\ \frac{4e^{t}\sin t}{e^2(e^2-1)}(\boldsymbol{i}-1) \end{pmatrix}$. Then, by simple calculation, when $t = \frac{5}{2}$, we gain $x(\frac{5}{2}, x_0) = \begin{pmatrix} 0 \\ 0 \end{pmatrix}$, which implies that (15) is state controllable. Thus, Theorem 3.4 holds.

The explicit solution $x = (x_1, x_2)^T$ of (15) is given as follows.

For $t \in [0, 1]$,

$$\begin{aligned} x(t) &= \varphi^{-1}\{e^{\zeta(A)t}\varphi(x_0)\} \\ &= \varphi^{-1}\left\{ \begin{pmatrix} e^{-t}\cos t & 0 & e^{-t}\sin t & 0 \\ 0 & e^{2t} & 0 & 0 \\ -e^{-t}\sin t & 0 & e^{-t}\cos t & 0 \\ 0 & 0 & 0 & e^{2t} \end{pmatrix} \begin{pmatrix} 1+\boldsymbol{i} \\ 1+\boldsymbol{i} \\ 0 \\ -1+\boldsymbol{i} \end{pmatrix} \right\} \\ &= \begin{pmatrix} e^{-t}\cos t + e^{-t}\cos t\boldsymbol{i} + e^{-t}\sin t\boldsymbol{j} - e^{-t}\sin t\boldsymbol{k} \\ e^{2t} + e^{2t}\boldsymbol{i} + e^{2t}\boldsymbol{j} + e^{2t}\boldsymbol{k} \end{pmatrix}. \end{aligned}$$

For $t \in (1, 2]$,

$$\begin{aligned} x(t) &= \varphi^{-1}\{e^{\zeta(A)(t-1)}\zeta((E+B))e^{\zeta(A)}\varphi(x_0) + \int_1^t e^{\zeta(A)(t-s)}\zeta(C)\varphi(u(s))ds\} \\ &= \varphi^{-1}\left\{ \begin{pmatrix} e^{-(t-1)}\cos(t-1) & 0 & e^{-(t-1)}\sin(t-1) & 0 \\ 0 & e^{2(t-1)} & 0 & 0 \\ -e^{-(t-1)}\sin(t-1) & 0 & e^{-(t-1)}\cos(t-1) & 0 \\ 0 & 0 & 0 & e^{2(t-1)} \end{pmatrix} \right. \end{aligned}$$

$$\times\begin{pmatrix}2&0&0&0\\0&0&0&\boldsymbol{i}\\0&0&2&0\\0&\boldsymbol{i}&0&0\end{pmatrix}\begin{pmatrix}e^{-1}\cos 1&0&e^{-1}\sin 1&0\\0&e^{2}&0&0\\-e^{-1}\sin 1&0&e^{-1}\cos 1&0\\0&0&0&e^{2}\end{pmatrix}\begin{pmatrix}1+\boldsymbol{i}\\1+\boldsymbol{i}\\0\\-1+\boldsymbol{i}\end{pmatrix}$$

$$+\int_1^t\begin{pmatrix}e^{-(t-s)}\cos(t-s)&0&e^{-(t-s)}\sin(t-s)&0\\0&e^{2(t-s)}&0&0\\-e^{-(t-s)}\sin(t-s)&0&e^{-(t-s)}\cos(t-s)&0\\0&0&0&e^{2(t-s)}\end{pmatrix}$$

$$\left.\times\begin{pmatrix}0&\boldsymbol{i}&0&0\\0&0&1&0\\0&0&0&-\boldsymbol{i}\\-1&0&0&0\end{pmatrix}\begin{pmatrix}\frac{4e^{-2s}}{e^{-4}(1-e^{-4})}(\boldsymbol{i}-1)\\\frac{4e^{s}\cos s}{e^{2}(e^{2}-1)}(\boldsymbol{i}-1)\\\frac{4e^{-2s}}{e^{-4}(1-e^{-4})}(\boldsymbol{i}+1)\\\frac{4e^{s}\sin s}{e^{2}(e^{2}-1)}(\boldsymbol{i}-1)\end{pmatrix}ds\right\}$$

$$=\begin{pmatrix}\frac{2e^{-t}(e^4-e^{2t})\cos t}{e^2(e^2-1)}+\frac{2e^{-t}(e^4-e^{2t})\cos t}{e^2(e^2-1)}\boldsymbol{i}+\frac{2e^{-t}(e^4-e^{2t})\sin t}{e^2(e^2-1)}\boldsymbol{j}-\frac{2e^{-t}(e^4-e^{2t})\sin t}{e^2(e^2-1)}\boldsymbol{k}\\\frac{e^{2t}(e^{-8}-e^{-4t})}{e^{-4}(1-e^{-4})}+\frac{e^{2t}(e^{-8}-e^{-4t})}{e^{-4}(1-e^{-4})}\boldsymbol{i}-\frac{e^{2t}(e^{-8}-e^{-4t})}{e^{-4}(1-e^{-4})}\boldsymbol{j}-\frac{e^{2t}(e^{-8}-e^{-4t})}{e^{-4}(1-e^{-4})}\boldsymbol{k}\end{pmatrix}.$$

For $t\in(2,\frac{5}{2}]$,

$$\begin{aligned}x(t)&=\varphi^{-1}\{e^{\zeta(A)(t-2)}\zeta((E+B))e^{\zeta(A)}\zeta((E+B))e^{\zeta(A)}\varphi(x_0)\\&\quad+e^{\zeta(A)(t-2)}\zeta((E+B))e^{\zeta(A)}\int_1^2 e^{\zeta(A)(1-s)}\zeta(C)\varphi(u(s))ds\}\end{aligned}$$

$$=\varphi^{-1}\left\{\begin{pmatrix}e^{-(t-2)}\cos(t-2)&0&e^{-(t-2)}\sin(t-2)&0\\0&e^{2(t-2)}&0&0\\-e^{-(t-2)}\sin(t-2)&0&e^{-(t-2)}\cos(t-2)&0\\0&0&0&e^{2(t-2)}\end{pmatrix}\begin{pmatrix}2&0&0&0\\0&0&0&\boldsymbol{i}\\0&0&2&0\\0&\boldsymbol{i}&0&0\end{pmatrix}\right.$$

$$\times\begin{pmatrix}e^{-1}\cos 1&0&e^{-1}\sin 1&0\\0&e^{2}&0&0\\-e^{-1}\sin 1&0&e^{-1}\cos 1&0\\0&0&0&e^{2}\end{pmatrix}\begin{pmatrix}2&0&0&0\\0&0&0&\boldsymbol{i}\\0&0&2&0\\0&\boldsymbol{i}&0&0\end{pmatrix}\begin{pmatrix}e^{-1}\cos 1&0&e^{-1}\sin 1&0\\0&e^{2}&0&0\\-e^{-1}\sin 1&0&e^{-1}\cos 1&0\\0&0&0&e^{2}\end{pmatrix}\begin{pmatrix}1+\boldsymbol{i}\\1+\boldsymbol{i}\\0\\-1+\boldsymbol{i}\end{pmatrix}$$

$$+\begin{pmatrix}e^{-(t-2)}\cos(t-2)&0&e^{-(t-2)}\sin(t-2)&0\\0&e^{2(t-2)}&0&0\\-e^{-(t-2)}\sin(t-2)&0&e^{-(t-2)}\cos(t-2)&0\\0&0&0&e^{2(t-2)}\end{pmatrix}\begin{pmatrix}2&0&0&0\\0&0&0&\boldsymbol{i}\\0&0&2&0\\0&\boldsymbol{i}&0&0\end{pmatrix}$$

$$\times\begin{pmatrix}e^{-1}\cos 1&0&e^{-1}\sin 1&0\\0&e^{2}&0&0\\-e^{-1}\sin 1&0&e^{-1}\cos 1&0\\0&0&0&e^{2}\end{pmatrix}\int_1^2\begin{pmatrix}e^{-(1-s)}\cos(1-s)&0&e^{-(1-s)}\sin(1-s)&0\\0&e^{2(1-s)}&0&0\\-e^{-(1-s)}\sin(1-s)&0&e^{-(1-s)}\cos(1-s)&0\\0&0&0&e^{2(1-s)}\end{pmatrix}$$

$$\left.\times\begin{pmatrix}0&\boldsymbol{i}&0&0\\0&0&1&0\\0&0&0&-\boldsymbol{i}\\-1&0&0&0\end{pmatrix}\begin{pmatrix}\frac{4e^{-2s}}{e^{-4}(1-e^{-4})}(\boldsymbol{i}-1)\\\frac{4e^{s}\cos s}{e^{2}(e^{2}-1)}(\boldsymbol{i}-1)\\\frac{4e^{-2s}}{e^{-4}(1-e^{-4})}(\boldsymbol{i}+1)\\\frac{4e^{s}\sin s}{e^{2}(e^{2}-1)}(\boldsymbol{i}-1)\end{pmatrix}ds\right\}$$

$$=\begin{pmatrix}0\\0\end{pmatrix}.$$

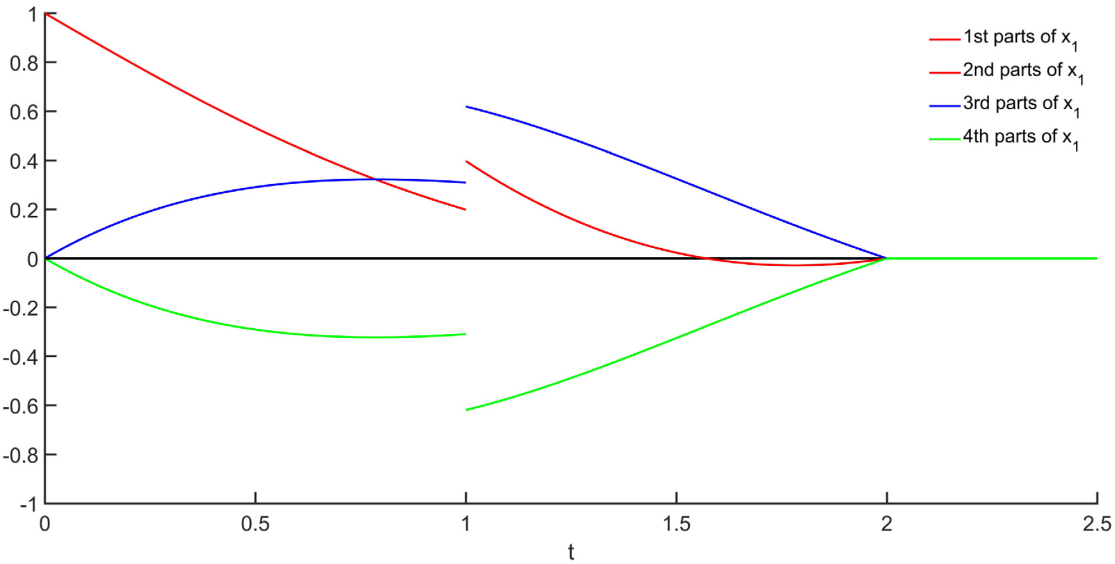

Fig. 1. Four parts of x_1 of (15).

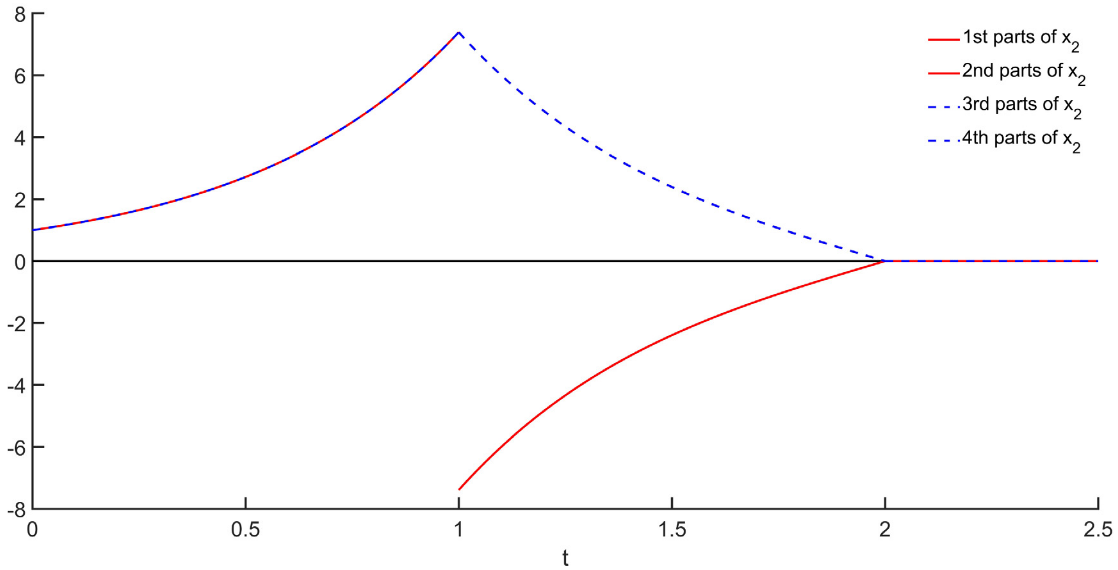

Fig. 2. Four parts of x_2 of (15).

The 1st parts of x_1 are $e^{-t}\cos t$, $\frac{2e^{-t}(e^4-e^{2t})\cos t}{e^2(e^2-1)}$, 0, respectively. See the red line in Fig. 1. The 2nd parts of x_1 are $e^{-t}\cos t$, $\frac{2e^{-t}(e^4-e^{2t})\cos t}{e^2(e^2-1)}$, 0, respectively. See the red line in Fig. 1. The 3rd parts of x_1 are $e^{-t}\sin t$, $\frac{2e^{-t}(e^4-e^{2t})\sin t}{e^2(e^2-1)}$, 0, respectively. See the blue line in Fig. 1. The 4th parts of x_1 are $-e^{-t}\sin t$, $-\frac{2e^{-t}(e^4-e^{2t})\sin t}{e^2(e^2-1)}$, 0, respectively. See the green line in Fig. 1.

The 1st parts of x_2 are e^{2t}, $\frac{e^{2t}(e^{-8}-e^{-4t})}{e^{-4}(1-e^{-4})}$, 0, respectively. See the red line in Fig. 2. The 2nd parts of x_2 are e^{2t}, $\frac{e^{2t}(e^{-8}-e^{-4t})}{e^{-4}(1-e^{-4})}$, 0, respectively. See the red line in Fig. 2. The 3rd parts of x_2 are e^{2t}, $-\frac{e^{2t}(e^{-8}-e^{-4t})}{e^{-4}(1-e^{-4})}$, 0, respectively. See the blue line in Fig. 2. The 4th parts of x_2 are e^{2t}, $-\frac{e^{2t}(e^{-8}-e^{-4t})}{e^{-4}(1-e^{-4})}$, 0, respectively. See the blue line in Fig. 2.

According to Fig. 1 shown, we observe that when $t = \frac{5}{2}$, 1st, 2nd, 3rd and 4th parts of x_1 are 0, respectively, which implies that $x_1(\frac{5}{2}) = 0$. According to Fig. 2 shown, we observe that when $t = \frac{5}{2}$, 1st, 2nd, 3rd and 4th parts of x_2 are 0, respectively, which implies that $x_2(\frac{5}{2}) = 0$. It is obvious that $x(\frac{5}{2}) = 0$. Hence, the system (15) is state controllable.

From (8),

$$
\begin{aligned}
\overline{U}_0 &= \int_0^1 e^{-\zeta(A)s}\zeta(C)\zeta(C^*)e^{-\zeta(A^*)s}ds \\
&= \int_0^1 \begin{pmatrix} e^s\cos s & 0 & -e^s\sin s & 0 \\ 0 & e^{-2s} & 0 & 0 \\ e^s\sin s & 0 & e^s\cos s & 0 \\ 0 & 0 & 0 & e^{-2s} \end{pmatrix} \begin{pmatrix} 0 & \boldsymbol{i} & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & -\boldsymbol{i} \\ -1 & 0 & 0 & 0 \end{pmatrix} \\
&\quad \times \begin{pmatrix} 0 & 0 & 0 & -1 \\ -\boldsymbol{i} & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & \boldsymbol{i} & 0 \end{pmatrix} \begin{pmatrix} e^s\cos s & 0 & e^s\sin s & 0 \\ 0 & e^{-2s} & 0 & 0 \\ -e^s\sin s & 0 & e^s\cos s & 0 \\ 0 & 0 & 0 & e^{-2s} \end{pmatrix} ds \\
&= \begin{pmatrix} \frac{1}{2}(e^2-1) & 0 & 0 & 0 \\ 0 & \frac{1}{4}(1-e^{-4}) & 0 & 0 \\ 0 & 0 & \frac{1}{2}(e^2-1) & 0 \\ 0 & 0 & 0 & \frac{1}{4}(1-e^{-4}) \end{pmatrix}, \\
\overline{U}_1 &= \int_1^2 e^{-\zeta(A)}\zeta((E+B)^{-1})e^{\zeta(A)(1-s)}\zeta(C)\zeta(C^*)e^{\zeta(A^*)(1-s)}\zeta((E+B^*)^{-1})e^{-\zeta(A^*)}ds \\
&= \int_1^2 \begin{pmatrix} e\cos 1 & 0 & -e\sin 1 & 0 \\ 0 & e^{-2} & 0 & 0 \\ e\sin 1 & 0 & e\cos 1 & 0 \\ 0 & 0 & 0 & e^{-2} \end{pmatrix} \begin{pmatrix} 2 & 0 & 0 & 0 \\ 0 & 0 & 0 & \boldsymbol{i} \\ 0 & 0 & 2 & 0 \\ 0 & \boldsymbol{i} & 0 & 0 \end{pmatrix}^{-1} \\
&\quad \times \begin{pmatrix} e^{-(1-s)}\cos(1-s) & 0 & e^{-(1-s)}\sin(1-s) & 0 \\ 0 & e^{2(1-s)} & 0 & 0 \\ -e^{-(1-s)}\sin(1-s) & 0 & e^{-(1-s)}\cos(1-s) & 0 \\ 0 & 0 & 0 & e^{2(1-s)} \end{pmatrix} \begin{pmatrix} 0 & \boldsymbol{i} & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & -\boldsymbol{i} \\ -1 & 0 & 0 & 0 \end{pmatrix} \\
&\quad \times \begin{pmatrix} 0 & 0 & 0 & -1 \\ -\boldsymbol{i} & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & \boldsymbol{i} & 0 \end{pmatrix} \begin{pmatrix} e^{-(1-s)}\cos(1-s) & 0 & -e^{-(1-s)}\sin(1-s) & 0 \\ 0 & e^{2(1-s)} & 0 & 0 \\ e^{-(1-s)}\sin(1-s) & 0 & e^{-(1-s)}\cos(1-s) & 0 \\ 0 & 0 & 0 & e^{2(1-s)} \end{pmatrix} \\
&\quad \times \begin{pmatrix} 2 & 0 & 0 & 0 \\ 0 & 0 & 0 & -\boldsymbol{i} \\ 0 & 0 & 2 & 0 \\ 0 & -\boldsymbol{i} & 0 & 0 \end{pmatrix}^{-1} \begin{pmatrix} e\cos 1 & 0 & e\sin 1 & 0 \\ 0 & e^{-2} & 0 & 0 \\ -e\sin 1 & 0 & e\cos 1 & 0 \\ 0 & 0 & 0 & e^{-2} \end{pmatrix} ds \\
&= \begin{pmatrix} \frac{1}{8}(e^4-e^2) & 0 & 0 & 0 \\ 0 & \frac{1}{4}e^{-4}(1-e^{-4}) & 0 & 0 \\ 0 & 0 & \frac{1}{8}(e^4-e^2) & 0 \\ 0 & 0 & 0 & \frac{1}{4}e^{-4}(1-e^{-4}) \end{pmatrix}, \\
\overline{U}_2 &= \int_2^{\frac{5}{2}} e^{-\zeta(A)}\zeta((E+B)^{-1})e^{-\zeta(A)}\zeta((E+B)^{-1})e^{\zeta(A)(2-s)}\zeta(C) \\
&\quad \zeta(C^*)e^{\zeta(A^*)(2-s)}\zeta((E+B^*)^{-1})e^{-\zeta(A^*)}\zeta((E+B^*)^{-1})e^{-\zeta(A^*)}ds \\
&= \int_2^{\frac{5}{2}} \begin{pmatrix} e\cos 1 & 0 & -e\sin 1 & 0 \\ 0 & e^{-2} & 0 & 0 \\ e\sin 1 & 0 & e\cos 1 & 0 \\ 0 & 0 & 0 & e^{-2} \end{pmatrix} \begin{pmatrix} 2 & 0 & 0 & 0 \\ 0 & 0 & 0 & \boldsymbol{i} \\ 0 & 0 & 2 & 0 \\ 0 & \boldsymbol{i} & 0 & 0 \end{pmatrix}^{-1} \begin{pmatrix} e\cos 1 & 0 & -e\sin 1 & 0 \\ 0 & e^{-2} & 0 & 0 \\ e\sin 1 & 0 & e\cos 1 & 0 \\ 0 & 0 & 0 & e^{-2} \end{pmatrix}
\end{aligned}
$$

$$
\begin{aligned}
&\times \begin{pmatrix} 2 & 0 & 0 & 0 \\ 0 & 0 & 0 & \boldsymbol{i} \\ 0 & 0 & 2 & 0 \\ 0 & \boldsymbol{i} & 0 & 0 \end{pmatrix}^{-1} \begin{pmatrix} e^{-(2-s)}\cos(2-s) & 0 & e^{-(2-s)}\sin(2-s) & 0 \\ 0 & e^{2(2-s)} & 0 & 0 \\ -e^{-(2-s)}\sin(2-s) & 0 & e^{-(2-s)}\cos(2-s) & 0 \\ 0 & 0 & 0 & e^{2(2-s)} \end{pmatrix} \begin{pmatrix} 0 & \boldsymbol{i} & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & -\boldsymbol{i} \\ -1 & 0 & 0 & 0 \end{pmatrix} \\
&\times \begin{pmatrix} 0 & 0 & 0 & -1 \\ -\boldsymbol{i} & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & \boldsymbol{i} & 0 \end{pmatrix} \begin{pmatrix} e^{-(2-s)}\cos(2-s) & 0 & -e^{-(2-s)}\sin(2-s) & 0 \\ 0 & e^{2(2-s)} & 0 & 0 \\ e^{-(2-s)}\sin(2-s) & 0 & e^{-(2-s)}\cos(2-s) & 0 \\ 0 & 0 & 0 & e^{2(2-s)} \end{pmatrix} \begin{pmatrix} 2 & 0 & 0 & 0 \\ 0 & 0 & 0 & -\boldsymbol{i} \\ 0 & 0 & 2 & 0 \\ 0 & -\boldsymbol{i} & 0 & 0 \end{pmatrix}^{-1} \\
&\times \begin{pmatrix} e\cos 1 & 0 & e\sin 1 & 0 \\ 0 & e^{-2} & 0 & 0 \\ -e\sin 1 & 0 & e\cos 1 & 0 \\ 0 & 0 & 0 & e^{-2} \end{pmatrix} \begin{pmatrix} 2 & 0 & 0 & 0 \\ 0 & 0 & 0 & -\boldsymbol{i} \\ 0 & 0 & 2 & 0 \\ 0 & -\boldsymbol{i} & 0 & 0 \end{pmatrix}^{-1} \begin{pmatrix} e\cos 1 & 0 & e\sin 1 & 0 \\ 0 & e^{-2} & 0 & 0 \\ -e\sin 1 & 0 & e\cos 1 & 0 \\ 0 & 0 & 0 & e^{-2} \end{pmatrix} ds \\
&= \begin{pmatrix} \frac{1}{32}e^4(e-1) & 0 & 0 & 0 \\ 0 & \frac{1}{4}e^{-8}(1-e^{-2}) & 0 & 0 \\ 0 & 0 & \frac{1}{32}e^4(e-1) & 0 \\ 0 & 0 & 0 & \frac{1}{4}e^{-8}(1-e^{-2}) \end{pmatrix}, \\
&\quad rank\,(\overline{U}_0, \overline{U}_1, \overline{U}_2) = 4,
\end{aligned}
$$

therefore, Theorem 3.6 holds.

$$
\zeta(C) = \begin{pmatrix} 0 & \boldsymbol{i} & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & -\boldsymbol{i} \\ -1 & 0 & 0 & 0 \end{pmatrix}, \quad \zeta(A)\zeta(C) = \begin{pmatrix} 0 & -\boldsymbol{i} & 0 & -\boldsymbol{i} \\ 0 & 0 & 2 & 0 \\ 0 & -\boldsymbol{i} & 0 & \boldsymbol{i} \\ -2 & 0 & 0 & 0 \end{pmatrix},
$$

$$
\zeta(A^2)\zeta(C) = \begin{pmatrix} 0 & 0 & 0 & 2\boldsymbol{i} \\ 0 & 0 & 4 & 0 \\ 0 & 2\boldsymbol{i} & 0 & 0 \\ -4 & 0 & 0 & 0 \end{pmatrix}, \quad \zeta(A^3)\zeta(C) = \begin{pmatrix} 0 & 2\boldsymbol{i} & 0 & -2\boldsymbol{i} \\ 0 & 0 & 8 & 0 \\ 0 & -2\boldsymbol{i} & 0 & -2\boldsymbol{i} \\ -8 & 0 & 0 & 0 \end{pmatrix},
$$

$$
rank\,(\zeta(C), \zeta(A)\zeta(C), \zeta(A^2)\zeta(C), \zeta(A^3)\zeta(C)) = 4,
$$

hence, Theorem 3.10 holds.

From Theorem 4.2,

$$
y(t) = \begin{cases} \varphi^{-1}\{\zeta(D)e^{\zeta(A)t}\varphi(x_0)\}, \ t \in [0, 1], \\ \varphi^{-1}\{\zeta(D)e^{\zeta(A)(t-1)}\zeta((E+B))e^{\zeta(A)}\varphi(x_0)\}, \ t \in (1, 2], \\ \varphi^{-1}\{\zeta(D)e^{\zeta(A)(t-2)}\zeta((E+B))e^{\zeta(A)}\zeta((E+B))e^{\zeta(A)}\varphi(x_0)\}, \ t \in (2, \frac{5}{2}]. \end{cases}
$$

The explicit solution $y = (y_1, y_2)^T$ of (15) is given as follows.

For $t \in [0, 1]$,

$$
\begin{aligned}
y(t) &= \varphi^{-1}\{\zeta(D)e^{\zeta(A)t}\varphi(x_0)\} \\
&= \varphi^{-1}\left\{ \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & \boldsymbol{i} \\ 0 & 0 & 1 & 0 \\ 0 & \boldsymbol{i} & 0 & 0 \end{pmatrix} \begin{pmatrix} e^{-t}\cos t & 0 & e^{-t}\sin t & 0 \\ 0 & e^{2t} & 0 & 0 \\ -e^{-t}\sin t & 0 & e^{-t}\cos t & 0 \\ 0 & 0 & 0 & e^{2t} \end{pmatrix} \begin{pmatrix} 1+\boldsymbol{i} \\ 1+\boldsymbol{i} \\ 0 \\ -1+\boldsymbol{i} \end{pmatrix} \right\} \\
&= \begin{pmatrix} e^{-t}\cos t + e^{-t}\cos t\boldsymbol{i} + e^{-t}\sin t\boldsymbol{j} - e^{-t}\sin t\boldsymbol{k} \\ -e^{2t} - e^{2t}\boldsymbol{i} + e^{2t}\boldsymbol{j} + e^{2t}\boldsymbol{k} \end{pmatrix}.
\end{aligned}
$$

For $t \in (1, 2]$,

$$
y(t) = \varphi^{-1}\{\zeta(D)e^{\zeta(A)(t-1)}\zeta((E+B))e^{\zeta(A)}\varphi(x_0)\}
$$

$$
= \varphi^{-1}\left\{\begin{pmatrix}1&0&0&0\\0&0&0&\boldsymbol{i}\\0&0&1&0\\0&\boldsymbol{i}&0&0\end{pmatrix}\begin{pmatrix}e^{-(t-1)}\cos(t-1)&0&e^{-(t-1)}\sin(t-1)&0\\0&e^{2(t-1)}&0&0\\-e^{-(t-1)}\sin(t-1)&0&e^{-(t-1)}\cos(t-1)&0\\0&0&0&e^{2(t-1)}\end{pmatrix}\right.
$$

$$
\left.\times\begin{pmatrix}2&0&0&0\\0&0&0&\boldsymbol{i}\\0&0&2&0\\0&\boldsymbol{i}&0&0\end{pmatrix}\begin{pmatrix}e^{-1}\cos 1&0&e^{-1}\sin 1&0\\0&e^{2}&0&0\\-e^{-1}\sin 1&0&e^{-1}\cos 1&0\\0&0&0&e^{2}\end{pmatrix}\begin{pmatrix}1+\boldsymbol{i}\\1+\boldsymbol{i}\\0\\-1+\boldsymbol{i}\end{pmatrix}\right\}
$$

$$
= \begin{pmatrix}2e^{-t}\cos t+2e^{-t}\cos t\boldsymbol{i}+2e^{-t}\sin t\boldsymbol{j}-2e^{-t}\sin t\boldsymbol{k}\\-e^{2t}-e^{2t}\boldsymbol{i}-e^{2t}\boldsymbol{j}-e^{2t}\boldsymbol{k}\end{pmatrix}.
$$

For $t \in (2, \frac{5}{2}]$,

$$
y(t) = \varphi^{-1}\{\zeta(D)e^{\zeta(A)(t-2)}\zeta((E+B))e^{\zeta(A)}\zeta((E+B))e^{\zeta(A)}\varphi(x_0)\}
$$

$$
= \varphi^{-1}\left\{\begin{pmatrix}1&0&0&0\\0&0&0&\boldsymbol{i}\\0&0&1&0\\0&\boldsymbol{i}&0&0\end{pmatrix}\begin{pmatrix}e^{-(t-2)}\cos(t-2)&0&e^{-(t-2)}\sin(t-2)&0\\0&e^{2(t-2)}&0&0\\-e^{-(t-2)}\sin(t-2)&0&e^{-(t-2)}\cos(t-2)&0\\0&0&0&e^{2(t-2)}\end{pmatrix}\right.
$$

$$
\times\begin{pmatrix}2&0&0&0\\0&0&0&\boldsymbol{i}\\0&0&2&0\\0&\boldsymbol{i}&0&0\end{pmatrix}\begin{pmatrix}e^{-1}\cos 1&0&e^{-1}\sin 1&0\\0&e^{2}&0&0\\-e^{-1}\sin 1&0&e^{-1}\cos 1&0\\0&0&0&e^{2}\end{pmatrix}
$$

$$
\left.\times\begin{pmatrix}2&0&0&0\\0&0&0&\boldsymbol{i}\\0&0&2&0\\0&\boldsymbol{i}&0&0\end{pmatrix}\begin{pmatrix}e^{-1}\cos 1&0&e^{-1}\sin 1&0\\0&e^{2}&0&0\\-e^{-1}\sin 1&0&e^{-1}\cos 1&0\\0&0&0&e^{2}\end{pmatrix}\begin{pmatrix}1+\boldsymbol{i}\\1+\boldsymbol{i}\\0\\-1+\boldsymbol{i}\end{pmatrix}\right\}
$$

$$
= \begin{pmatrix}4e^{-t}\cos t+4e^{-t}\cos t\boldsymbol{i}+4e^{-t}\sin t\boldsymbol{j}-4e^{-t}\sin t\boldsymbol{k}\\e^{2t}+e^{2t}\boldsymbol{i}-e^{2t}\boldsymbol{j}-e^{2t}\boldsymbol{k}\end{pmatrix}.
$$

For the initial value $x_0 = \begin{pmatrix}1+\boldsymbol{i}\\1+\boldsymbol{i}+\boldsymbol{j}+\boldsymbol{k}\end{pmatrix}$ given,

for $t \in [0, 1]$,

$$
y(t) = \varphi^{-1}\{\zeta(D)e^{\zeta(A)t}\}x_0 = \begin{pmatrix}e^{-t}\cos t+e^{-t}\cos t\boldsymbol{i}+e^{-t}\sin t\boldsymbol{j}-e^{-t}\sin t\boldsymbol{k}\\-e^{2t}-e^{2t}\boldsymbol{i}+e^{2t}\boldsymbol{j}+e^{2t}\boldsymbol{k}\end{pmatrix}.
$$

for $t \in (1, 2]$,

$$
y(t) = \varphi^{-1}\{\zeta(D)e^{\zeta(A)(t-1)}\zeta((E+B))e^{\zeta(A)}\}x_0 = \begin{pmatrix}2e^{-t}\cos t+2e^{-t}\cos t\boldsymbol{i}+2e^{-t}\sin t\boldsymbol{j}-2e^{-t}\sin t\boldsymbol{k}\\-e^{2t}-e^{2t}\boldsymbol{i}-e^{2t}\boldsymbol{j}-e^{2t}\boldsymbol{k}\end{pmatrix}.
$$

for $t \in (2, \frac{5}{2}]$,

$$
y(t) = \varphi^{-1}\{\zeta(D)e^{\zeta(A)(t-2)}\zeta((E+B))e^{\zeta(A)}\zeta((E+B))e^{\zeta(A)}\}x_0 = \begin{pmatrix}4e^{-t}\cos t+4e^{-t}\cos t\boldsymbol{i}+4e^{-t}\sin t\boldsymbol{j}-4e^{-t}\sin t\boldsymbol{k}\\e^{2t}+e^{2t}\boldsymbol{i}-e^{2t}\boldsymbol{j}-e^{2t}\boldsymbol{k}\end{pmatrix},
$$

we observe that for $t \in [0, 1]$, $t \in (1, 2]$, $t \in (2, \frac{5}{2}]$, the initial state x_0 is uniquely determined by the corresponding system output y, respectively, which implies that the system (15) is state observable.

The 1st parts of y_1 are $e^{-t}\cos t$, $2e^{-t}\cos t$, $4e^{-t}\cos t$, respectively. See the red line in Fig. 3. The 2nd parts of y_1 are $e^{-t}\cos t$, $2e^{-t}\cos t$, $4e^{-t}\cos t$, respectively. See the red line in Fig. 3. The 3rd parts of y_1 are $e^{-t}\sin t$, $2e^{-t}\sin t$, $4e^{-t}\sin t$, respectively. See the blue line in Fig. 3. The 4th parts of y_1 are $-e^{-t}\sin t$, $-2e^{-t}\sin t$, $-4e^{-t}\sin t$, respectively. See the green line in Fig. 3.

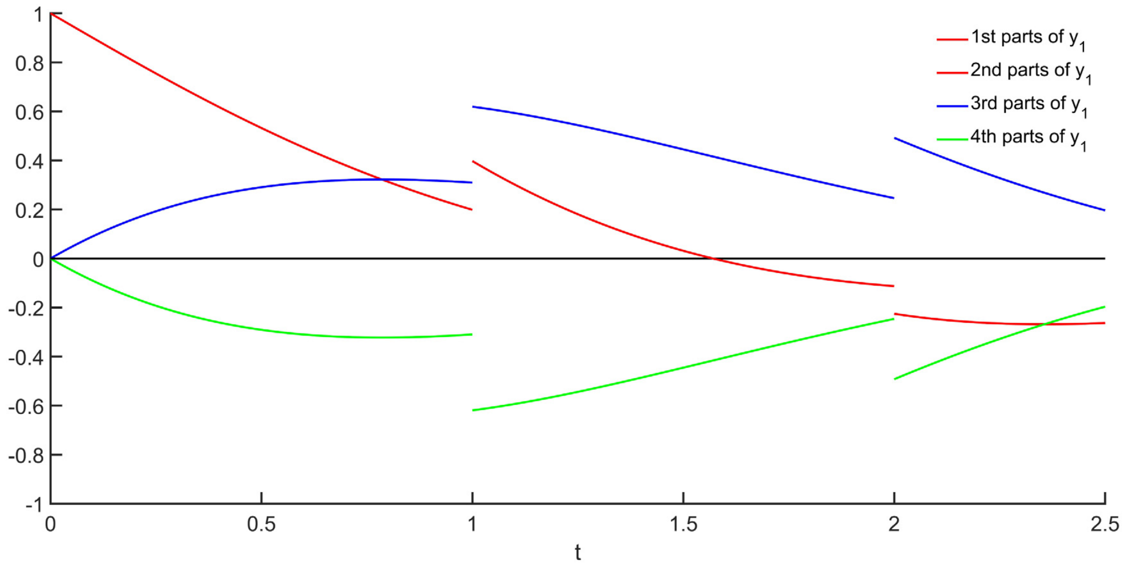

Fig. 3. Four parts of y_1 of (15).

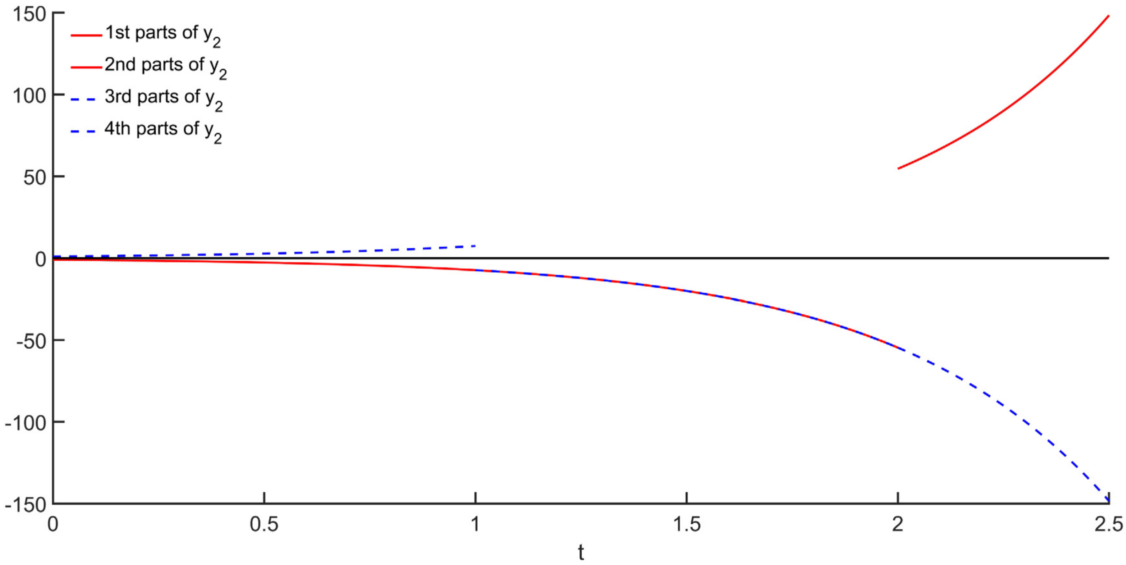

Fig. 4. Four parts of y_2 of (15).

The 1st parts of y_2 are $-e^{2t}, -e^{2t}, e^{2t}$, respectively. See the red line in Fig. 4. The 2nd parts of y_2 are $-e^{2t}, -e^{2t}, e^{2t}$, respectively. See the red line in Fig. 4. The 3rd parts of y_2 are $e^{2t}, -e^{2t}, -e^{2t}$, respectively. See the blue line in Fig. 4. The 4th parts of y_2 are $e^{2t}, -e^{2t}, -e^{2t}$, respectively. See the blue line in Fig. 4.

From (12),

$$\begin{aligned}
\overline{N}(t_0,t_1) &= \int_0^1 e^{\zeta(A^*)s}\zeta(D^*)\zeta(D)e^{\zeta(A)s}ds \\
&= \int_0^1 \begin{pmatrix} e^{-s}\cos s & 0 & -e^{-s}\sin s & 0 \\ 0 & e^{2s} & 0 & 0 \\ e^{-s}\sin s & 0 & e^{-s}\cos s & 0 \\ 0 & 0 & 0 & e^{2s} \end{pmatrix} \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & -\boldsymbol{i} \\ 0 & 0 & 1 & 0 \\ 0 & -\boldsymbol{i} & 0 & 0 \end{pmatrix}
\end{aligned}$$

$$\times \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & \boldsymbol{i} \\ 0 & 0 & 1 & 0 \\ 0 & \boldsymbol{i} & 0 & 0 \end{pmatrix} \begin{pmatrix} e^{-s}\cos s & 0 & e^{-s}\sin s & 0 \\ 0 & e^{2s} & 0 & 0 \\ -e^{-s}\sin s & 0 & e^{-s}\cos s & 0 \\ 0 & 0 & 0 & e^{2s} \end{pmatrix} ds$$

$$= \begin{pmatrix} \frac{1}{2}(1-e^{-2}) & 0 & 0 & 0 \\ 0 & \frac{1}{4}(e^{4}-1) & 0 & 0 \\ 0 & 0 & \frac{1}{2}(1-e^{-2}) & 0 \\ 0 & 0 & 0 & \frac{1}{4}(e^{4}-1) \end{pmatrix},$$

$$\overline{N}(t_1,t_2) = \int_1^2 e^{\zeta(A^*)}\zeta((E+B^*))e^{\zeta(A^*)(s-1)}\zeta(D^*)\zeta(D)e^{\zeta(A)(s-1)}\zeta((E+B))e^{\zeta(A)}ds$$

$$= \int_1^2 \begin{pmatrix} e^{-1}\cos 1 & 0 & -e^{-1}\sin 1 & 0 \\ 0 & e^{2} & 0 & 0 \\ e^{-1}\sin 1 & 0 & e^{-1}\cos 1 & 0 \\ 0 & 0 & 0 & e^{2} \end{pmatrix} \begin{pmatrix} 2 & 0 & 0 & 0 \\ 0 & 0 & 0 & -\boldsymbol{i} \\ 0 & 0 & 2 & 0 \\ 0 & -\boldsymbol{i} & 0 & 0 \end{pmatrix}$$

$$\times \begin{pmatrix} e^{-(s-1)}\cos(s-1) & 0 & -e^{-(s-1)}\sin(s-1) & 0 \\ 0 & e^{2(s-1)} & 0 & 0 \\ e^{-(s-1)}\sin(s-1) & 0 & e^{-(s-1)}\cos(s-1) & 0 \\ 0 & 0 & 0 & e^{2(s-1)} \end{pmatrix} \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & -\boldsymbol{i} \\ 0 & 0 & 1 & 0 \\ 0 & -\boldsymbol{i} & 0 & 0 \end{pmatrix}$$

$$\times \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & \boldsymbol{i} \\ 0 & 0 & 1 & 0 \\ 0 & \boldsymbol{i} & 0 & 0 \end{pmatrix} \begin{pmatrix} e^{-(s-1)}\cos(s-1) & 0 & e^{-(s-1)}\sin(s-1) & 0 \\ 0 & e^{2(s-1)} & 0 & 0 \\ -e^{-(s-1)}\sin(s-1) & 0 & e^{-(s-1)}\cos(s-1) & 0 \\ 0 & 0 & 0 & e^{2(s-1)} \end{pmatrix}$$

$$\times \begin{pmatrix} 2 & 0 & 0 & 0 \\ 0 & 0 & 0 & \boldsymbol{i} \\ 0 & 0 & 2 & 0 \\ 0 & \boldsymbol{i} & 0 & 0 \end{pmatrix} \begin{pmatrix} e^{-1}\cos 1 & 0 & e^{-1}\sin 1 & 0 \\ 0 & e^{2} & 0 & 0 \\ -e^{-1}\sin 1 & 0 & e^{-1}\cos 1 & 0 \\ 0 & 0 & 0 & e^{2} \end{pmatrix} ds$$

$$= \begin{pmatrix} 2e^{-2}(1-e^{-2}) & 0 & 0 & 0 \\ 0 & \frac{1}{4}e^{4}(e^{4}-1) & 0 & 0 \\ 0 & 0 & 2e^{-2}(1-e^{-2}) & 0 \\ 0 & 0 & 0 & \frac{1}{4}e^{4}(e^{4}-1) \end{pmatrix},$$

$$\overline{N}(t_2,t_l) = \int_2^{\frac{5}{2}} e^{\zeta(A^*)}\zeta((E+B^*))e^{\zeta(A^*)}\zeta((E+B^*))e^{\zeta(A^*)(s-2)}\zeta(D^*)$$
$$\zeta(D)e^{\zeta(A)(s-2)}\zeta((E+B))e^{\zeta(A)}\zeta((E+B))e^{\zeta(A)}ds$$

$$= \int_2^{\frac{5}{2}} \begin{pmatrix} e^{-1}\cos 1 & 0 & -e^{-1}\sin 1 & 0 \\ 0 & e^{2} & 0 & 0 \\ e^{-1}\sin 1 & 0 & e^{-1}\cos 1 & 0 \\ 0 & 0 & 0 & e^{2} \end{pmatrix} \begin{pmatrix} 2 & 0 & 0 & 0 \\ 0 & 0 & 0 & -\boldsymbol{i} \\ 0 & 0 & 2 & 0 \\ 0 & -\boldsymbol{i} & 0 & 0 \end{pmatrix}$$

$$\times \begin{pmatrix} e^{-1}\cos 1 & 0 & -e^{-1}\sin 1 & 0 \\ 0 & e^{2} & 0 & 0 \\ e^{-1}\sin 1 & 0 & e^{-1}\cos 1 & 0 \\ 0 & 0 & 0 & e^{2} \end{pmatrix} \begin{pmatrix} 2 & 0 & 0 & 0 \\ 0 & 0 & 0 & -\boldsymbol{i} \\ 0 & 0 & 2 & 0 \\ 0 & -\boldsymbol{i} & 0 & 0 \end{pmatrix}$$

$$\times \begin{pmatrix} e^{-(s-2)}\cos(s-2) & 0 & -e^{-(s-2)}\sin(s-2) & 0 \\ 0 & e^{2(s-2)} & 0 & 0 \\ e^{-(s-2)}\sin(s-2) & 0 & e^{-(s-2)}\cos(s-2) & 0 \\ 0 & 0 & 0 & e^{2(s-2)} \end{pmatrix} \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & -\boldsymbol{i} \\ 0 & 0 & 1 & 0 \\ 0 & -\boldsymbol{i} & 0 & 0 \end{pmatrix}$$

$$
\begin{aligned}
&\times \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & \boldsymbol{i} \\ 0 & 0 & 1 & 0 \\ 0 & \boldsymbol{i} & 0 & 0 \end{pmatrix} \begin{pmatrix} e^{-(s-2)}\cos(s-2) & 0 & e^{-(s-2)}\sin(s-2) & 0 \\ 0 & e^{2(s-2)} & 0 & 0 \\ -e^{-(s-2)}\sin(s-2) & 0 & e^{-(s-2)}\cos(s-2) & 0 \\ 0 & 0 & 0 & e^{2(s-2)} \end{pmatrix} \\
&\times \begin{pmatrix} 2 & 0 & 0 & 0 \\ 0 & 0 & 0 & \boldsymbol{i} \\ 0 & 0 & 2 & 0 \\ 0 & \boldsymbol{i} & 0 & 0 \end{pmatrix} \begin{pmatrix} e^{-1}\cos 1 & 0 & e^{-1}\sin 1 & 0 \\ 0 & e^{2} & 0 & 0 \\ -e^{-1}\sin 1 & 0 & e^{-1}\cos 1 & 0 \\ 0 & 0 & 0 & e^{2} \end{pmatrix} \\
&\times \begin{pmatrix} 2 & 0 & 0 & 0 \\ 0 & 0 & 0 & \boldsymbol{i} \\ 0 & 0 & 2 & 0 \\ 0 & \boldsymbol{i} & 0 & 0 \end{pmatrix} \begin{pmatrix} e^{-1}\cos 1 & 0 & e^{-1}\sin 1 & 0 \\ 0 & e^{2} & 0 & 0 \\ -e^{-1}\sin 1 & 0 & e^{-1}\cos 1 & 0 \\ 0 & 0 & 0 & e^{2} \end{pmatrix} ds \\
&= \begin{pmatrix} 8e^{-4}(1-e^{-1}) & 0 & 0 & 0 \\ 0 & \frac{1}{4}e^{8}(e^{2}-1) & 0 & 0 \\ 0 & 0 & 8e^{-4}(1-e^{-1}) & 0 \\ 0 & 0 & 0 & \frac{1}{4}e^{8}(e^{2}-1) \end{pmatrix},
\end{aligned}
$$

$$
\begin{aligned}
\overline{N}(t_0, t_l) &= \overline{N}(t_0, t_1) + \overline{N}(t_1, t_2) + \overline{N}(t_2, t_l) \\
&= \begin{pmatrix} \frac{1}{2}+\frac{3}{2}e^{-2}+6e^{-4}-8e^{-5} & 0 & 0 & 0 \\ 0 & -\frac{1}{4}+\frac{1}{4}e^{10} & 0 & 0 \\ 0 & 0 & \frac{1}{2}+\frac{3}{2}e^{-2}+6e^{-4}-8e^{-5} & 0 \\ 0 & 0 & 0 & -\frac{1}{4}+\frac{1}{4}e^{10} \end{pmatrix}, \\
&\quad rank\ \overline{N}(t_0, t_l) = 4,
\end{aligned}
$$

therefore, Theorem 4.2 holds.

$$
\zeta(D) = \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & \boldsymbol{i} \\ 0 & 0 & 1 & 0 \\ 0 & \boldsymbol{i} & 0 & 0 \end{pmatrix}, \quad \zeta(D)\zeta(A) = \begin{pmatrix} -1 & 0 & 1 & 0 \\ 0 & 0 & 0 & 2\boldsymbol{i} \\ -1 & 0 & -1 & 0 \\ 0 & 2\boldsymbol{i} & 0 & 0 \end{pmatrix},
$$

$$
\zeta(D)\zeta(A^2) = \begin{pmatrix} 0 & 0 & -2 & 0 \\ 0 & 0 & 0 & 4\boldsymbol{i} \\ 2 & 0 & 0 & 0 \\ 0 & 4\boldsymbol{i} & 0 & 0 \end{pmatrix}, \quad \zeta(D)\zeta(A^3) = \begin{pmatrix} 2 & 0 & 2 & 0 \\ 0 & 0 & 0 & 8\boldsymbol{i} \\ -2 & 0 & 2 & 0 \\ 0 & 8\boldsymbol{i} & 0 & 0 \end{pmatrix},
$$

$$
rank \begin{pmatrix} \zeta(D) \\ \zeta(D)\zeta(A) \\ \zeta(D)\zeta(A^2) \\ \zeta(D)\zeta(A^3) \end{pmatrix} = 4,
$$

hence, Theorem 4.5 holds.

Example 5.2. Considering linear QIDEs:

$$
\begin{cases}
\dot{x}(t) = \begin{pmatrix} -1+\boldsymbol{j} & 0 \\ 0 & 2 \end{pmatrix} x(t) + \begin{pmatrix} 0 & \boldsymbol{i} \\ \boldsymbol{j} & 0 \end{pmatrix} u(t),\ t \in J' := [0, \frac{5}{2}]\backslash\{t_k\},\ k = 1, 2, \\
\Delta x(t_k) = \begin{pmatrix} 1 & 0 \\ 0 & -1+\boldsymbol{k} \end{pmatrix} x(t_k^-),\ k = 1, 2, \\
y(t) = \begin{pmatrix} 1 & 0 \\ 0 & \boldsymbol{k} \end{pmatrix} x(t), \\
x(0) = \begin{pmatrix} 1+\boldsymbol{i} \\ 1+\boldsymbol{i}+\boldsymbol{j}+\boldsymbol{k} \end{pmatrix},
\end{cases} \tag{16}
$$

where $t_k = k$.

We have

$$A=\begin{pmatrix}-1+\boldsymbol{j} & 0\\ 0 & 2\end{pmatrix},\ B=\begin{pmatrix}1 & 0\\ 0 & -1+\boldsymbol{k}\end{pmatrix},\ C=\begin{pmatrix}0 & \boldsymbol{i}\\ \boldsymbol{j} & 0\end{pmatrix},\ D=\begin{pmatrix}1 & 0\\ 0 & \boldsymbol{k}\end{pmatrix},$$

$$x_0=x(0)=\begin{pmatrix}1+\boldsymbol{i}\\ 1+\boldsymbol{i}+\boldsymbol{j}+\boldsymbol{k}\end{pmatrix}.$$

From (8),

$$\begin{aligned}V_1 &= V(t_1,t_2)\\ &=\int_1^2 e^{A(1-s)}CC^*e^{A^*(1-s)}ds\\ &=\int_1^2 \begin{pmatrix}e^{(-1+\boldsymbol{j})(1-s)} & 0\\ 0 & e^{2(1-s)}\end{pmatrix}\begin{pmatrix}0 & \boldsymbol{i}\\ \boldsymbol{j} & 0\end{pmatrix}\begin{pmatrix}0 & -\boldsymbol{j}\\ -\boldsymbol{i} & 0\end{pmatrix}\begin{pmatrix}e^{(-1-\boldsymbol{j})(1-s)} & 0\\ 0 & e^{2(1-s)}\end{pmatrix}ds\\ &=\begin{pmatrix}\frac{1}{2}(e^2-1) & 0\\ 0 & \frac{1}{4}(1-e^{-4})\end{pmatrix},\\ &\quad rank\ V_1=2.\end{aligned}$$

From (18),

$$\begin{aligned}u(t) &= -C^*e^{A^*(1-t)}V^{-1}(t_1,t_2)(E+B)e^Ax_0\\ &=-\begin{pmatrix}0 & -\boldsymbol{j}\\ -\boldsymbol{i} & 0\end{pmatrix}\begin{pmatrix}e^{(-1-\boldsymbol{j})(1-t)} & 0\\ 0 & e^{2(1-t)}\end{pmatrix}\begin{pmatrix}\frac{2}{e^2-1} & 0\\ 0 & \frac{4}{1-e^{-4}}\end{pmatrix}\begin{pmatrix}2 & 0\\ 0 & \boldsymbol{k}\end{pmatrix}\begin{pmatrix}e^{-1+\boldsymbol{j}} & 0\\ 0 & e^2\end{pmatrix}\begin{pmatrix}1+\boldsymbol{i}\\ 1+\boldsymbol{i}+\boldsymbol{j}+\boldsymbol{k}\end{pmatrix}\\ &=\begin{pmatrix}\frac{4e^{-2t}}{e^{-4}(1-e^{-4})}(-1+\boldsymbol{i}-\boldsymbol{j}+\boldsymbol{k})\\ \frac{4e^{(1-\boldsymbol{j})t}}{e^2(e^2-1)}(-1+\boldsymbol{i})\end{pmatrix},\end{aligned}$$

where $t\in(1,2]$.

When $t_l=\frac{5}{2}$,

$$\begin{aligned}x(t_l,x_0) &= e^{\frac{1}{2}A}(E+B)e^A(E+B)e^Ax_0+e^{\frac{1}{2}A}(E+B)e^A\int_1^2 e^{A(1-s)}Cu(s)ds\\ &=\begin{pmatrix}e^{\frac{-1+\boldsymbol{j}}{2}} & 0\\ 0 & e\end{pmatrix}\begin{pmatrix}2 & 0\\ 0 & \boldsymbol{k}\end{pmatrix}\begin{pmatrix}e^{-1+\boldsymbol{j}} & 0\\ 0 & e^2\end{pmatrix}\begin{pmatrix}2 & 0\\ 0 & \boldsymbol{k}\end{pmatrix}\begin{pmatrix}e^{-1+\boldsymbol{j}} & 0\\ 0 & e^2\end{pmatrix}\begin{pmatrix}1+\boldsymbol{i}\\ 1+\boldsymbol{i}+\boldsymbol{j}+\boldsymbol{k}\end{pmatrix}\\ &\quad+\begin{pmatrix}e^{\frac{-1+\boldsymbol{j}}{2}} & 0\\ 0 & e\end{pmatrix}\begin{pmatrix}2 & 0\\ 0 & \boldsymbol{k}\end{pmatrix}\begin{pmatrix}e^{-1+\boldsymbol{j}} & 0\\ 0 & e^2\end{pmatrix}\\ &\quad\times\int_1^2\begin{pmatrix}e^{(-1+\boldsymbol{j})(1-s)} & 0\\ 0 & e^{2(1-s)}\end{pmatrix}\begin{pmatrix}0 & \boldsymbol{i}\\ \boldsymbol{j} & 0\end{pmatrix}\begin{pmatrix}\frac{4e^{-2s}}{e^{-4}(1-e^{-4})}(-1+\boldsymbol{i}-\boldsymbol{j}+\boldsymbol{k})\\ \frac{4e^{(1-\boldsymbol{j})s}}{e^2(e^2-1)}(-1+\boldsymbol{i})\end{pmatrix}ds\\ &=\begin{pmatrix}0\\ 0\end{pmatrix},\end{aligned}$$

then (16) is state controllable. Thus, Theorem 3.5 holds.

From (8),

$$\begin{aligned}U_0 &= \int_0^1 e^{-As}CC^*e^{-A^*s}ds\\ &=\int_0^1\begin{pmatrix}e^{(1-\boldsymbol{j})s} & 0\\ 0 & e^{-2s}\end{pmatrix}\begin{pmatrix}0 & \boldsymbol{i}\\ \boldsymbol{j} & 0\end{pmatrix}\begin{pmatrix}0 & -\boldsymbol{j}\\ -\boldsymbol{i} & 0\end{pmatrix}\begin{pmatrix}e^{(1+\boldsymbol{j})s} & 0\\ 0 & e^{-2s}\end{pmatrix}ds\\ &=\begin{pmatrix}\frac{1}{2}(e^2-1) & 0\\ 0 & \frac{1}{4}(1-e^{-4})\end{pmatrix},\\ U_1 &= \int_1^2 e^{-A}(E+B)^{-1}e^{A(1-s)}CC^*e^{A^*(1-s)}(E+B^*)^{-1}e^{-A^*}ds\\ &=\int_1^2\begin{pmatrix}e^{1-\boldsymbol{j}} & 0\\ 0 & e^{-2}\end{pmatrix}\begin{pmatrix}2 & 0\\ 0 & \boldsymbol{k}\end{pmatrix}^{-1}\begin{pmatrix}e^{(-1+\boldsymbol{j})(1-s)} & 0\\ 0 & e^{2(1-s)}\end{pmatrix}\begin{pmatrix}0 & \boldsymbol{i}\\ \boldsymbol{j} & 0\end{pmatrix}\end{aligned}$$

$$
\begin{aligned}
&\times \begin{pmatrix} 0 & -\boldsymbol{j} \\ -\boldsymbol{i} & 0 \end{pmatrix} \begin{pmatrix} e^{(-1-\boldsymbol{j})(1-s)} & 0 \\ 0 & e^{2(1-s)} \end{pmatrix} \begin{pmatrix} 2 & 0 \\ 0 & -\boldsymbol{k} \end{pmatrix}^{-1} \begin{pmatrix} e^{1+\boldsymbol{j}} & 0 \\ 0 & e^{-2} \end{pmatrix} ds \\
&= \begin{pmatrix} \frac{1}{8}e^2(e^2-1) & 0 \\ 0 & \frac{1}{4}e^{-4}(1-e^{-4}) \end{pmatrix},
\end{aligned}
$$

$$
\begin{aligned}
U_2 &= \int_2^{\frac{5}{2}} e^{-A}(E+B)^{-1}e^{-A}(E+B)^{-1}e^{A(2-s)}CC^*e^{A^*(2-s)}(E+B^*)^{-1}e^{-A^*}(E+B^*)^{-1}e^{-A^*}ds \\
&= \int_2^{\frac{5}{2}} \begin{pmatrix} e^{1-\boldsymbol{j}} & 0 \\ 0 & e^{-2} \end{pmatrix} \begin{pmatrix} 2 & 0 \\ 0 & \boldsymbol{k} \end{pmatrix}^{-1} \begin{pmatrix} e^{1-\boldsymbol{j}} & 0 \\ 0 & e^{-2} \end{pmatrix} \begin{pmatrix} 2 & 0 \\ 0 & \boldsymbol{k} \end{pmatrix}^{-1} \begin{pmatrix} e^{(-1+\boldsymbol{j})(2-s)} & 0 \\ 0 & e^{2(2-s)} \end{pmatrix} \begin{pmatrix} 0 & \boldsymbol{i} \\ \boldsymbol{j} & 0 \end{pmatrix} \\
&\quad \times \begin{pmatrix} 0 & -\boldsymbol{j} \\ -\boldsymbol{i} & 0 \end{pmatrix} \begin{pmatrix} e^{(-1-\boldsymbol{j})(2-s)} & 0 \\ 0 & e^{2(2-s)} \end{pmatrix} \begin{pmatrix} 2 & 0 \\ 0 & -\boldsymbol{k} \end{pmatrix}^{-1} \begin{pmatrix} e^{1+\boldsymbol{j}} & 0 \\ 0 & e^{-2} \end{pmatrix} \begin{pmatrix} 2 & 0 \\ 0 & -\boldsymbol{k} \end{pmatrix}^{-1} \begin{pmatrix} e^{1+\boldsymbol{j}} & 0 \\ 0 & e^{-2} \end{pmatrix} ds \\
&= \begin{pmatrix} \frac{1}{32}e^4(e-1) & 0 \\ 0 & \frac{1}{4}e^{-8}(1-e^{-2}) \end{pmatrix},
\end{aligned}
$$

$$
rank\,(U_0, U_1, U_2) = 2,
$$

therefore, Theorem 3.7 holds.

$$
C = \begin{pmatrix} 0 & \boldsymbol{i} \\ \boldsymbol{j} & 0 \end{pmatrix}, \ AC = \begin{pmatrix} 0 & -\boldsymbol{i}-\boldsymbol{k} \\ 2\boldsymbol{j} & 0 \end{pmatrix}, \ A^2C = \begin{pmatrix} 0 & 2\boldsymbol{k} \\ 4\boldsymbol{j} & 0 \end{pmatrix}, \ A^3C = \begin{pmatrix} 0 & -2\boldsymbol{k}+2\boldsymbol{i} \\ 8\boldsymbol{j} & 0 \end{pmatrix},
$$

$$
rank\,(C, AC, A^2C, A^3C) = 2,
$$

hence, Theorem 3.11 holds.

From Theorem 4.3,

$$
y(t) = \begin{cases} De^{At}x_0, \ t \in [0,1], \\ De^{A(t-1)}(E+B)e^Ax_0, \ t \in (1,2], \\ De^{A(t-2)}(E+B)e^A(E+B)e^Ax_0, \ t \in (2, \dfrac{5}{2}]. \end{cases}
$$

The explicit solution $y = (y_1, y_2)^T$ of (16) is given as follows.

For $t \in [0, 1]$,

$$
\begin{aligned}
y(t) &= De^{At}x_0 \\
&= \begin{pmatrix} 1 & 0 \\ 0 & \boldsymbol{k} \end{pmatrix} \begin{pmatrix} e^{(-1+\boldsymbol{j})t} & 0 \\ 0 & e^{2t} \end{pmatrix} \begin{pmatrix} 1+\boldsymbol{i} \\ 1+\boldsymbol{i}+\boldsymbol{j}+\boldsymbol{k} \end{pmatrix} \\
&= \begin{pmatrix} e^{-t}\cos t + e^{-t}\cos t\boldsymbol{i} + e^{-t}\sin t\boldsymbol{j} - e^{-t}\sin t\boldsymbol{k} \\ -e^{2t} - e^{2t}\boldsymbol{i} + e^{2t}\boldsymbol{j} + e^{2t}\boldsymbol{k} \end{pmatrix}.
\end{aligned}
$$

For $t \in (1, 2]$,

$$
\begin{aligned}
y(t) &= De^{A(t-1)}(E+B)e^Ax_0 \\
&= \begin{pmatrix} 1 & 0 \\ 0 & \boldsymbol{k} \end{pmatrix} \begin{pmatrix} e^{(-1+\boldsymbol{j})(t-1)} & 0 \\ 0 & e^{2(t-1)} \end{pmatrix} \begin{pmatrix} 2 & 0 \\ 0 & \boldsymbol{k} \end{pmatrix} \begin{pmatrix} e^{-1+\boldsymbol{j}} & 0 \\ 0 & e^2 \end{pmatrix} \begin{pmatrix} 1+\boldsymbol{i} \\ 1+\boldsymbol{i}+\boldsymbol{j}+\boldsymbol{k} \end{pmatrix} \\
&= \begin{pmatrix} 2e^{-t}\cos t + 2e^{-t}\cos t\boldsymbol{i} + 2e^{-t}\sin t\boldsymbol{j} - 2e^{-t}\sin t\boldsymbol{k} \\ -e^{2t} - e^{2t}\boldsymbol{i} - e^{2t}\boldsymbol{j} - e^{2t}\boldsymbol{k} \end{pmatrix}.
\end{aligned}
$$

For $t \in (2, \frac{5}{2}]$,

$$
\begin{aligned}
y(t) &= De^{A(t-2)}(E+B)e^A(E+B)e^Ax_0 \\
&= \begin{pmatrix} 1 & 0 \\ 0 & \boldsymbol{k} \end{pmatrix} \begin{pmatrix} e^{(-1+\boldsymbol{j})(t-2)} & 0 \\ 0 & e^{2(t-2)} \end{pmatrix} \begin{pmatrix} 2 & 0 \\ 0 & \boldsymbol{k} \end{pmatrix} \begin{pmatrix} e^{-1+\boldsymbol{j}} & 0 \\ 0 & e^2 \end{pmatrix} \begin{pmatrix} 2 & 0 \\ 0 & \boldsymbol{k} \end{pmatrix} \begin{pmatrix} e^{-1+\boldsymbol{j}} & 0 \\ 0 & e^2 \end{pmatrix} \begin{pmatrix} 1+\boldsymbol{i} \\ 1+\boldsymbol{i}+\boldsymbol{j}+\boldsymbol{k} \end{pmatrix} \\
&= \begin{pmatrix} 4e^{-t}\cos t + 4e^{-t}\cos t\boldsymbol{i} + 4e^{-t}\sin t\boldsymbol{j} - 4e^{-t}\sin t\boldsymbol{k} \\ e^{2t} + e^{2t}\boldsymbol{i} - e^{2t}\boldsymbol{j} - e^{2t}\boldsymbol{k} \end{pmatrix}.
\end{aligned}
$$

Therefore, the system (16) is state observable.

From (12),

$$
\begin{aligned}
N(t_0,t_1) &= \int_0^1 e^{A^*s}D^*De^{As}ds\\
&= \int_0^1 \begin{pmatrix} e^{(-1-\boldsymbol{j})s} & 0\\ 0 & e^{2s}\end{pmatrix}\begin{pmatrix}1 & 0\\ 0 & -\boldsymbol{k}\end{pmatrix}\begin{pmatrix}1 & 0\\ 0 & \boldsymbol{k}\end{pmatrix}\begin{pmatrix} e^{(-1+\boldsymbol{j})s} & 0\\ 0 & e^{2s}\end{pmatrix}ds\\
&= \begin{pmatrix}\frac{1}{2}(1-e^{-2}) & 0\\ 0 & \frac{1}{4}(e^4-1)\end{pmatrix},\\
N(t_1,t_2) &= \int_1^2 e^{A^*}(E+B^*)e^{A^*(s-1)}D^*De^{A(s-1)}(E+B)e^{A}ds\\
&= \int_1^2 \begin{pmatrix} e^{-1-\boldsymbol{j}} & 0\\ 0 & e^{2}\end{pmatrix}\begin{pmatrix}2 & 0\\ 0 & -\boldsymbol{k}\end{pmatrix}\begin{pmatrix} e^{(-1-\boldsymbol{j})(s-1)} & 0\\ 0 & e^{2(s-1)}\end{pmatrix}\begin{pmatrix}1 & 0\\ 0 & -\boldsymbol{k}\end{pmatrix}\\
&\quad\times\begin{pmatrix}1 & 0\\ 0 & \boldsymbol{k}\end{pmatrix}\begin{pmatrix} e^{(-1+\boldsymbol{j})(s-1)} & 0\\ 0 & e^{2(s-1)}\end{pmatrix}\begin{pmatrix}2 & 0\\ 0 & \boldsymbol{k}\end{pmatrix}\begin{pmatrix} e^{-1+\boldsymbol{j}} & 0\\ 0 & e^{2}\end{pmatrix}ds\\
&= \begin{pmatrix}2e^{-2}(1-e^{-2}) & 0\\ 0 & \frac{1}{4}e^4(e^4-1)\end{pmatrix},\\
N(t_2,t_l) &= \int_2^{\frac{5}{2}} e^{A^*}(E+B^*)e^{A^*}(E+B^*)e^{A^*(s-2)}D^*De^{A(s-2)}(E+B)e^{A}(E+B)e^{A}ds\\
&= \int_2^{\frac{5}{2}} \begin{pmatrix} e^{-1-\boldsymbol{j}} & 0\\ 0 & e^{2}\end{pmatrix}\begin{pmatrix}2 & 0\\ 0 & -\boldsymbol{k}\end{pmatrix}\begin{pmatrix} e^{-1-\boldsymbol{j}} & 0\\ 0 & e^{2}\end{pmatrix}\begin{pmatrix}2 & 0\\ 0 & -\boldsymbol{k}\end{pmatrix}\begin{pmatrix} e^{(-1-\boldsymbol{j})(s-2)} & 0\\ 0 & e^{2(s-2)}\end{pmatrix}\begin{pmatrix}1 & 0\\ 0 & -\boldsymbol{k}\end{pmatrix}\\
&\quad\times\begin{pmatrix}1 & 0\\ 0 & \boldsymbol{k}\end{pmatrix}\begin{pmatrix} e^{(-1+\boldsymbol{j})(s-2)} & 0\\ 0 & e^{2(s-2)}\end{pmatrix}\begin{pmatrix}2 & 0\\ 0 & \boldsymbol{k}\end{pmatrix}\begin{pmatrix} e^{-1+\boldsymbol{j}} & 0\\ 0 & e^{2}\end{pmatrix}\begin{pmatrix}2 & 0\\ 0 & \boldsymbol{k}\end{pmatrix}\begin{pmatrix} e^{-1+\boldsymbol{j}} & 0\\ 0 & e^{2}\end{pmatrix}ds\\
&= \begin{pmatrix}8e^{-4}(1-e^{-1}) & 0\\ 0 & \frac{1}{4}e^8(e^2-1)\end{pmatrix},\\
N(t_0,t_l) &= N(t_0,t_1)+N(t_1,t_2)+N(t_2,t_l)\\
&= \begin{pmatrix}\frac{1}{2}+\frac{3}{2}e^{-2}+6e^{-4}-8e^{-5} & 0\\ 0 & -\frac{1}{4}+\frac{1}{4}e^{10}\end{pmatrix},\\
&\quad rank\, N(t_0,t_l)=2,
\end{aligned}
$$

therefore, Theorem 4.3 holds.

$$
D=\begin{pmatrix}1 & 0\\ 0 & \boldsymbol{k}\end{pmatrix},\ DA=\begin{pmatrix}-1+\boldsymbol{j} & 0\\ 0 & 2\boldsymbol{k}\end{pmatrix},\ DA^2=\begin{pmatrix}-2\boldsymbol{j} & 0\\ 0 & 4\boldsymbol{k}\end{pmatrix},\ DA^3=\begin{pmatrix}2\boldsymbol{j}+2 & 0\\ 0 & 8\boldsymbol{k}\end{pmatrix},
$$

$$
rank\begin{pmatrix}D\\ DA\\ DA^2\\ DA^3\end{pmatrix}=2,
$$

hence, Theorem 4.6 holds.

Conclusion

At the end of this paper, we conclude the main results, strengths and limitations of our study and provide comparisons with some previous results.

The controllability and observability for linear QIDEs are investigated. Several sufficient and necessary criteria for state controllability of the system are established in the sense of the complex-valued by Theorem 3.4, Theorem 3.6, Theorem 3.10, Theorem 3.12 and the quaternion-valued by Theorem 3.5, Theorem 3.7, Theorem 3.11, Theorem 3.13, respectively. At the same time, several sufficient and necessary criteria for state observability of the system are derived in the sense of the complex-valued by Theorem 4.2, Theorem 4.5, Theorem 4.7 and the quaternion-valued by Theorem 4.3, Theorem 4.6, Theorem 4.8, respectively.

By means of the adjoint matrix of quaternion matrix and the isomorphism between quaternion vector space and complex variables space, the n-dimension quaternion-valued system can be converted to the $2n$-dimension complex-valued system, which yields the equivalence relationship of each other. Therefore, one can search the solutions of the equivalent $2n$-dimension complex-valued system instead of the original n-dimension quaternion-valued system. This results extend the results in [27] to impulsive cases.

Hence, the mains results obtained for the controllability and observability for linear QIDEs (1) in the sense of the complex-valued and the quaternion-valued are equivalent to each other, which is a major innovation. We not only establish the relationship between the two, but also solve relevant problems in two approaches. In the forthcoming work, we will consider the controllability and observability for nonlinear QIDEs.

Owing to the non-exchangeability of quaternion multiplication, the assumed conditions $\zeta(A)\zeta(B) = \zeta(B)\zeta(A)$ and $AB = BA$ in Theorem 3.12, Theorem 3.13, Theorem 4.7 and Theorem 4.8 are extremely strong, which is the limitation of our study. However, without the assumed conditions $\zeta(A)\zeta(B) = \zeta(B)\zeta(A)$ and $AB = BA$, we will not be able to draw relevant conclusions.

The sufficient and necessary criteria for state controllability and state observability acquired in this paper are new and different from the previous conclusions in [33,43,45,47]. Suo et al. [33] only deduced the representations of the solutions for linear QIDEs in the sense of the complex-valued and quaternion-valued, respectively, and did not discuss the state controllability and state observability for the system. Guan et al. [43] only researched the controllability and observability for a class of linear real-valued impulsive systems, not QIDEs. In [43], the authors assumed that $\Delta x(t_k) = c_i x(t_i^-)$, where c_i are constants, $i = 1, 2, \ldots$, and the solution of the system studied can be expressed by

$$\begin{aligned}x(t) = &\prod_{j=1}^{k}(1+c_i)X(t,t_0)x_0 + \sum_{i=1}^{k}\prod_{j=1}^{k}(1+c_j)\int_{t_{i-1}}^{t_i}X(t,s)B(s)u(s)ds\\ &+\int_{t_k}^{t}X(t,s)B(s)u(s)ds,\end{aligned}$$

where $t \in (t_k, t_{k+1}]$, $k = 1, 2, \ldots$. Here, $(1 + c_i)$ can exchange with X, B and x_0, which makes the investigation of the controllability and observability of the system relatively simple, and please see the relevant proofs in [43]. However, owing to the non-exchangeability of quaternion multiplication, the research for QIDEs (1) is more complicated than the previous results, which is the biggest challenge and difficulty for us to study the controllability and observability of QIDEs (1). Chen et al. [45] just considered the controllability and observability for linear QDEs, which is without impulses. Recently, Suo et al. [47] studied the controllability and observability of linear QIDEs and nonlinear QIDEs via the represent of solution under the condition $AB = BA$, and the control is chosen in the entire interval $[t_0, t_l]$. In the current paper, we investigate the controllability and observability of linear QIDEs via the represent of solution under the condition $AB \neq BA$, and the control is chosen in a subinterval $(t_f, t_{f+1}] \subset [t_0, t_l], f \in \{0, 1, \ldots, k-1\}$. Thus the current proof and technique are different from [47].

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

No data was used for the research described in the article

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Appendix A. Proof of Theorem 3.4

Proof. First, we assume that $\exists\, f \in \{0, 1, \ldots, k-1\}$ s.t. $\text{rank}(\overline{V}_f) = 2n$, which means that $\overline{V}(t_f, t_{f+1})$ is invertible. Then giving a $\varphi(x_0) \in \mathbb{C}^{2n\times 1}$, we choose

$$\varphi(u(t)) = \begin{cases} -\zeta(C^*)e^{\zeta(A^*)(t_f-t)}\overline{V}^{-1}(t_f,t_{f+1})\prod_{j=f}^{1}\left[\zeta((E+B))e^{\zeta(A)(t_j-t_{j-1})}\right]\varphi(x_0), \\ t\in(t_f,t_{f+1}], \\ 0, \\ t\in[t_0,t_l]\backslash(t_f,t_{f+1}]. \end{cases} \tag{17}$$

According to (2), (8) and (17), the corresponding solution of (1) with $x(t_0) = x_0$ can be expressed as

$$
\begin{aligned}
& x(t_l, x_0) \\
&= \varphi^{-1}\Big\{ e^{\zeta(A)(t_l-t_k)} \prod_{j=k}^{1} \big[\zeta((E+B)) e^{\zeta(A)(t_j-t_{j-1})}\big] \varphi(x_0) \\
&\quad + e^{\zeta(A)(t_l-t_k)} \sum_{i=1}^{k} \prod_{j=k}^{i} \big[\zeta((E+B)) e^{\zeta(A)(t_j-t_{j-1})}\big] \int_{t_{i-1}}^{t_i} e^{\zeta(A)(t_{i-1}-s)} \zeta(C) \varphi(u(s)) ds \\
&\quad + \int_{t_k}^{t_l} e^{\zeta(A)(t_l-s)} \zeta(C) \varphi(u(s)) ds \Big\} \\
&= \varphi^{-1}\Big\{ e^{\zeta(A)(t_l-t_k)} \prod_{j=k}^{1} \big[\zeta((E+B)) e^{\zeta(A)(t_j-t_{j-1})}\big] \varphi(x_0) \\
&\quad + e^{\zeta(A)(t_l-t_k)} \prod_{j=k}^{f+1} \big[\zeta((E+B)) e^{\zeta(A)(t_j-t_{j-1})}\big] \int_{t_f}^{t_{f+1}} e^{\zeta(A)(t_f-s)} \zeta(C) \varphi(u(s)) ds \Big\} \\
&= \varphi^{-1}\Big\{ e^{\zeta(A)(t_l-t_k)} \prod_{j=k}^{1} \big[\zeta((E+B)) e^{\zeta(A)(t_j-t_{j-1})}\big] \varphi(x_0) \\
&\quad - e^{\zeta(A)(t_l-t_k)} \prod_{j=k}^{f+1} \big[\zeta((E+B)) e^{\zeta(A)(t_j-t_{j-1})}\big] \int_{t_f}^{t_{f+1}} e^{\zeta(A)(t_f-s)} \zeta(C) \\
&\quad \times \zeta(C^*) e^{\zeta(A^*)(t_f-s)} \overline{V}^{-1}(t_f, t_{f+1}) \prod_{j=f}^{1} \big[\zeta((E+B)) e^{\zeta(A)(t_j-t_{j-1})}\big] \varphi(x_0) ds \Big\} \\
&= \varphi^{-1}\Big\{ e^{\zeta(A)(t_l-t_k)} \prod_{j=k}^{1} \big[\zeta((E+B)) e^{\zeta(A)(t_j-t_{j-1})}\big] \Big[\varphi(x_0) \\
&\quad - \prod_{j=1}^{f} \big[e^{-\zeta(A)(t_j-t_{j-1})} \zeta((E+B)^{-1})\big] \int_{t_f}^{t_{f+1}} e^{\zeta(A)(t_f-s)} \zeta(C) \\
&\quad \times \zeta(C^*) e^{\zeta(A^*)(t_f-s)} \overline{V}^{-1}(t_f, t_{f+1}) ds \prod_{j=f}^{1} \big[\zeta((E+B)) e^{\zeta(A)(t_j-t_{j-1})}\big] \varphi(x_0) \Big] \Big\} \\
&= \varphi^{-1}\Big\{ e^{\zeta(A)(t_l-t_k)} \prod_{j=k}^{1} \big[\zeta((E+B)) e^{\zeta(A)(t_j-t_{j-1})}\big] \Big[\varphi(x_0) \\
&\quad - \prod_{j=1}^{f} \big[e^{-\zeta(A)(t_j-t_{j-1})} \zeta((E+B)^{-1})\big] \overline{V}(t_f, t_{f+1}) \\
&\quad \times \overline{V}^{-1}(t_f, t_{f+1}) \prod_{j=f}^{1} \big[\zeta((E+B)) e^{\zeta(A)(t_j-t_{j-1})}\big] \varphi(x_0) \Big] \Big\} \\
&= \varphi^{-1}\Big\{ e^{\zeta(A)(t_l-t_k)} \prod_{j=k}^{1} \big[\zeta((E+B)) e^{\zeta(A)(t_j-t_{j-1})}\big] \Big[\varphi(x_0) \\
&\quad - \prod_{j=1}^{f} \big[e^{-\zeta(A)(t_j-t_{j-1})} \zeta((E+B)^{-1})\big] \prod_{j=f}^{1} \big[\zeta((E+B)) e^{\zeta(A)(t_j-t_{j-1})}\big] \varphi(x_0) \Big] \Big\} \\
&= \varphi^{-1}\Big\{ e^{\zeta(A)(t_l-t_k)} \prod_{j=k}^{1} \big[\zeta((E+B)) e^{\zeta(A)(t_j-t_{j-1})}\big] \big[\varphi(x_0) - \varphi(x_0)\big] \Big\} \\
&= 0.
\end{aligned}
$$

Thus, (1) is state controllable on $[t_0, t_l]$.
The proof is completed. □

Appendix B. Proof of Theorem 3.5

Proof. First, we assume that $\exists\, f \in \{0, 1, \ldots, k-1\}$ s.t. $\operatorname{rank}(V_f) = n$, which means that $V(t_f, t_{f+1})$ is invertible. Then giving a $x_0 \in \mathbb{H}^{n\times 1}$, we choose

$$u(t) = \begin{cases} -C^* e^{A^*(t_f - t)} V^{-1}(t_f, t_{f+1}) \prod_{j=f}^{1} \left[(E+B)e^{A(t_j - t_{j-1})}\right] x_0, \\ t \in (t_f, t_{f+1}], \\ 0, \\ t \in [t_0, t_l] \backslash (t_f, t_{f+1}]. \end{cases} \tag{18}$$

According to (3), (8) and (18), the corresponding solution of (1) with $x(t_0) = x_0$ can be expressed as

$$\begin{aligned}
& x(t_l, x_0) \\
&= e^{A(t_l - t_k)} \prod_{j=k}^{1} \left[(E+B)e^{A(t_j - t_{j-1})}\right] x_0 \\
&\quad + e^{A(t_l - t_k)} \sum_{i=1}^{k} \prod_{j=k}^{i} \left[(E+B)e^{A(t_j - t_{j-1})}\right] \int_{t_{i-1}}^{t_i} e^{A(t_{i-1} - s)} Cu(s)ds \\
&\quad + \int_{t_k}^{t_l} e^{A(t_l - s)} Cu(s)ds \\
&= e^{A(t_l - t_k)} \prod_{j=k}^{1} \left[(E+B)e^{A(t_j - t_{j-1})}\right] x_0 \\
&\quad + e^{A(t_l - t_k)} \prod_{j=k}^{f+1} \left[(E+B)e^{A(t_j - t_{j-1})}\right] \int_{t_f}^{t_{f+1}} e^{A(t_f - s)} Cu(s)ds \\
&= e^{A(t_l - t_k)} \prod_{j=k}^{1} \left[(E+B)e^{A(t_j - t_{j-1})}\right] x_0 \\
&\quad - e^{A(t_l - t_k)} \prod_{j=k}^{f+1} \left[(E+B)e^{A(t_j - t_{j-1})}\right] \int_{t_f}^{t_{f+1}} e^{A(t_f - s)} C \\
&\quad \times C^* e^{A^*(t_f - s)} V^{-1}(t_f, t_{f+1}) \prod_{j=f}^{1} \left[(E+B)e^{A(t_j - t_{j-1})}\right] x_0 ds \\
&= e^{A(t_l - t_k)} \prod_{j=k}^{1} \left[(E+B)e^{A(t_j - t_{j-1})}\right] \Bigg[x_0 - \prod_{j=1}^{f} \left[e^{-A(t_j - t_{j-1})}(E+B)^{-1}\right] \int_{t_f}^{t_{f+1}} e^{A(t_f - s)} C \\
&\quad \times C^* e^{A^*(t_f - s)} V^{-1}(t_f, t_{f+1}) ds \prod_{j=f}^{1} \left[(E+B)e^{A(t_j - t_{j-1})}\right] x_0 \Bigg] \\
&= e^{A(t_l - t_k)} \prod_{j=k}^{1} \left[(E+B)e^{A(t_j - t_{j-1})}\right] \Bigg[x_0 - \prod_{j=1}^{f} \left[e^{-A(t_j - t_{j-1})}(E+B)^{-1}\right] V(t_f, t_{f+1}) \\
&\quad \times V^{-1}(t_f, t_{f+1}) \prod_{j=f}^{1} \left[(E+B)e^{A(t_j - t_{j-1})}\right] x_0 \Bigg] \\
&= e^{A(t_l - t_k)} \prod_{j=k}^{1} \left[(E+B)e^{A(t_j - t_{j-1})}\right] \Bigg[x_0 - \prod_{j=1}^{f} \left[e^{-A(t_j - t_{j-1})}(E+B)^{-1}\right] \\
&\quad \times \prod_{j=f}^{1} \left[(E+B)e^{A(t_j - t_{j-1})}\right] x_0 \Bigg]
\end{aligned}$$

$$= e^{A(t_l-t_k)}\prod_{j=k}^{1}\big[(E+B)e^{A(t_j-t_{j-1})}\big]\big[x_0-x_0\big]$$
$$= 0.$$

Thus, (1) is state controllable on $[t_0, t_l]$.
The proof is completed. □

Appendix C. Proof of Theorem 3.6

Proof. Assume that (1) is state controllable on $[t_0, t_l]$ and $\mathrm{rank}(\overline{U}_0, \overline{U}_1, \ldots, \overline{U}_k) < 2n$.
Then, $\exists\ \varphi(x_\alpha) \neq 0 \in \mathbb{C}^{2n\times 1}$ s.t.

$$\begin{aligned}
0 &= \varphi(x_\alpha^*)\overline{U}(t_0, t_1)\varphi(x_\alpha)\\
&= \int_{t_0}^{t_1} \varphi(x_\alpha^*)e^{\zeta(A)(t_0-s)}\zeta(C)\zeta(C^*)e^{\zeta(A^*)(t_0-s)}\varphi(x_\alpha)ds\\
&= \int_{t_0}^{t_1}\Big[\varphi(x_\alpha^*)e^{\zeta(A)(t_0-s)}\zeta(C)\Big]\Big[\varphi(x_\alpha^*)e^{\zeta(A)(t_0-s)}\zeta(C)\Big]^* ds,\\
0 &= \varphi(x_\alpha^*)\overline{U}(t_{i-1}, t_i)\varphi(x_\alpha)\\
&= \int_{t_{i-1}}^{t_i} \varphi(x_\alpha^*)\prod_{j=1}^{i-1}\big[e^{-\zeta(A)(t_j-t_{j-1})}\zeta((E+B)^{-1})\big]e^{\zeta(A)(t_{i-1}-s)}\zeta(C)\\
&\quad\times \zeta(C^*)e^{\zeta(A^*)(t_{i-1}-s)}\prod_{j=1}^{i-1}\big[\zeta((E+B^*)^{-1})e^{-\zeta(A^*)(t_j-t_{j-1})}\big]\varphi(x_\alpha)ds\\
&= \int_{t_{i-1}}^{t_i}\Big[\varphi(x_\alpha^*)\prod_{j=1}^{i-1}\big[e^{-\zeta(A)(t_j-t_{j-1})}\zeta((E+B)^{-1})\big]e^{\zeta(A)(t_{i-1}-s)}\zeta(C)\Big]\\
&\quad\times\Big[\varphi(x_\alpha^*)\prod_{j=1}^{i-1}\big[e^{-\zeta(A)(t_j-t_{j-1})}\zeta((E+B)^{-1})\big]e^{\zeta(A)(t_{i-1}-s)}\zeta(C)\Big]^* ds,\\
&\quad i = 2, \ldots, k,\\
0 &= \varphi(x_\alpha^*)\overline{U}(t_k, t_l)\varphi(x_\alpha)\\
&= \int_{t_k}^{t_l} \varphi(x_\alpha^*)\prod_{j=1}^{k}\big[e^{-\zeta(A)(t_j-t_{j-1})}\zeta((E+B)^{-1})\big]e^{\zeta(A)(t_k-s)}\zeta(C)\\
&\quad\times \zeta(C^*)e^{\zeta(A^*)(t_k-s)}\prod_{j=1}^{k}\big[\zeta((E+B^*)^{-1})e^{-\zeta(A^*)(t_j-t_{j-1})}\big]\varphi(x_\alpha)ds\\
&= \int_{t_k}^{t_l}\Big[\varphi(x_\alpha^*)\prod_{j=1}^{k}\big[e^{-\zeta(A)(t_j-t_{j-1})}\zeta((E+B)^{-1})\big]e^{\zeta(A)(t_k-s)}\zeta(C)\Big]\\
&\quad\times\Big[\varphi(x_\alpha^*)\prod_{j=1}^{k}\big[e^{-\zeta(A)(t_j-t_{j-1})}\zeta((E+B)^{-1})\big]e^{\zeta(A)(t_k-s)}\zeta(C)\Big]^* ds. \qquad (19)
\end{aligned}$$

According to (19), we obtain

$$\begin{cases}
\varphi(x_\alpha^*)e^{\zeta(A)(t_0-t)}\zeta(C) = 0,\ \ t\in[t_0, t_1],\\
\varphi(x_\alpha^*)\prod_{j=1}^{i-1}\big[e^{-\zeta(A)(t_j-t_{j-1})}\zeta((E+B)^{-1})\big]e^{\zeta(A)(t_{i-1}-t)}\zeta(C) = 0,\ \ t\in(t_{i-1}, t_i],\ \ i=2,\ldots,k,\\
\varphi(x_\alpha^*)\prod_{j=1}^{k}\big[e^{-\zeta(A)(t_j-t_{j-1})}\zeta((E+B)^{-1})\big]e^{\zeta(A)(t_k-t)}\zeta(C) = 0,\ \ t\in(t_k, t_l].
\end{cases} \qquad (20)$$

Because (1) is state controllable on $[t_0, t_l]$, then there is a piece continuous control function $\varphi(u(t))$ for initial state $\varphi(x_0) = \varphi(x_\alpha)$ such that

$$
\begin{aligned}
0 &= x(t_l, x_\alpha) \\
&= \varphi^{-1}\Bigg\{ e^{\zeta(A)(t_l-t_k)} \prod_{j=k}^{1} \big[\zeta((E+B)) e^{\zeta(A)(t_j-t_{j-1})}\big] \varphi(x_\alpha) \\
&\quad + e^{\zeta(A)(t_l-t_k)} \sum_{i=1}^{k} \prod_{j=k}^{i} \big[\zeta((E+B)) e^{\zeta(A)(t_j-t_{j-1})}\big] \int_{t_{i-1}}^{t_i} e^{\zeta(A)(t_{i-1}-s)} \zeta(C) \varphi(u(s)) ds \\
&\quad + \int_{t_k}^{t_l} e^{\zeta(A)(t_l-s)} \zeta(C) \varphi(u(s)) ds \Bigg\} \\
&= \varphi^{-1}\Bigg\{ e^{\zeta(A)(t_l-t_k)} \prod_{j=k}^{1} \big[\zeta((E+B)) e^{\zeta(A)(t_j-t_{j-1})}\big] \varphi(x_\alpha) \\
&\quad + e^{\zeta(A)(t_l-t_k)} \prod_{j=k}^{1} \big[\zeta((E+B)) e^{\zeta(A)(t_j-t_{j-1})}\big] \int_{t_0}^{t_1} e^{\zeta(A)(t_0-s)} \zeta(C) \varphi(u(s)) ds \\
&\quad + e^{\zeta(A)(t_l-t_k)} \sum_{i=2}^{k} \prod_{j=k}^{i} \big[\zeta((E+B)) e^{\zeta(A)(t_j-t_{j-1})}\big] \int_{t_{i-1}}^{t_i} e^{\zeta(A)(t_{i-1}-s)} \zeta(C) \varphi(u(s)) ds \\
&\quad + \int_{t_k}^{t_l} e^{\zeta(A)(t_l-s)} \zeta(C) \varphi(u(s)) ds \Bigg\}.
\end{aligned} \tag{21}
$$

From (21), we derive

$$
\begin{aligned}
\varphi(x_\alpha) &= -\int_{t_0}^{t_1} e^{\zeta(A)(t_0-s)} \zeta(C) \varphi(u(s)) ds \\
&\quad - \sum_{i=2}^{k} \prod_{j=1}^{i-1} \big[e^{-\zeta(A)(t_j-t_{j-1})} \zeta((E+B)^{-1})\big] \int_{t_{i-1}}^{t_i} e^{\zeta(A)(t_{i-1}-s)} \zeta(C) \varphi(u(s)) ds \\
&\quad - \prod_{j=1}^{k} \big[e^{-\zeta(A)(t_j-t_{j-1})} \zeta((E+B)^{-1})\big] \int_{t_k}^{t_l} e^{\zeta(A)(t_k-s)} \zeta(C) \varphi(u(s)) ds \Bigg\}.
\end{aligned} \tag{22}
$$

By multiplying $\varphi(x_\alpha^*)$ to both sides of (22) from left, from (20), we obtain

$$
\begin{aligned}
\varphi(x_\alpha^*)\varphi(x_\alpha) &= -\int_{t_0}^{t_1} \varphi(x_\alpha^*) e^{\zeta(A)(t_0-s)} \zeta(C) \varphi(u(s)) ds \\
&\quad - \sum_{i=2}^{k} \varphi(x_\alpha^*) \prod_{j=1}^{i-1} \big[e^{-\zeta(A)(t_j-t_{j-1})} \zeta((E+B)^{-1})\big] \int_{t_{i-1}}^{t_i} e^{\zeta(A)(t_{i-1}-s)} \zeta(C) \varphi(u(s)) ds \\
&\quad - \varphi(x_\alpha^*) \prod_{j=1}^{k} \big[e^{-\zeta(A)(t_j-t_{j-1})} \zeta((E+B)^{-1})\big] \int_{t_k}^{t_l} e^{\zeta(A)(t_k-s)} \zeta(C) \varphi(u(s)) ds \Bigg\} \\
&= 0.
\end{aligned}
$$

This contradicts $\varphi(x_\alpha) \neq 0$ and we can acquire that $rank(\overline{U}_0\ \overline{U}_1\ \cdots\ \overline{U}_k) = 2n$.
The proof is completed. □

Appendix D. Proof of Theorem 3.7

Proof. Assume that (1) is state controllable on $[t_0, t_l]$ and $\text{rank}(U_0, U_1, \dots, U_k) < n$.
Then, $\exists\, x_\alpha \neq 0 \in \mathbb{H}^{n\times 1}$ s.t.

$$
\begin{aligned}
0 &= x_\alpha^* U(t_0, t_1) x_\alpha \\
&= \int_{t_0}^{t_1} x_\alpha^* e^{A(t_0-s)} C C^* e^{A^*(t_0-s)} x_\alpha ds \\
&= \int_{t_0}^{t_1} \Big[x_\alpha^* e^{A(t_0-s)} C\Big] \Big[x_\alpha^* e^{A(t_0-s)} C\Big]^* ds,
\end{aligned}
$$

$$
\begin{aligned}
0 &= x_\alpha^* U(t_{i-1}, t_i) x_\alpha \\
&= \int_{t_{i-1}}^{t_i} x_\alpha^* \prod_{j=1}^{i-1} \left[e^{-A(t_j - t_{j-1})} (E+B)^{-1} \right] e^{A(t_{i-1}-s)} C \\
&\quad \times C^* e^{A^*(t_{i-1}-s)} \prod_{j=1}^{i-1} \left[(E+B^*)^{-1} e^{-A^*(t_j - t_{j-1})} \right] x_\alpha ds \\
&= \int_{t_{i-1}}^{t_i} \left[x_\alpha^* \prod_{j=1}^{i-1} \left[e^{-A(t_j - t_{j-1})} (E+B)^{-1} \right] e^{A(t_{i-1}-s)} C \right] \\
&\quad \times \left[x_\alpha^* \prod_{j=1}^{i-1} \left[e^{-A(t_j - t_{j-1})} (E+B)^{-1} \right] e^{A(t_{i-1}-s)} C \right]^* ds, \\
&\quad i = 2, \ldots, k, \\
0 &= x_\alpha^* U(t_k, t_l) x_\alpha \\
&= \int_{t_k}^{t_l} x_\alpha^* \prod_{j=1}^{k} \left[e^{-A(t_j - t_{j-1})} (E+B)^{-1} \right] e^{A(t_k-s)} C \\
&\quad \times C^* e^{A^*(t_k-s)} \prod_{j=1}^{k} \left[(E+B^*)^{-1} e^{-A^*(t_j - t_{j-1})} \right] x_\alpha ds \\
&= \int_{t_k}^{t_l} \left[x_\alpha^* \prod_{j=1}^{k} \left[e^{-A(t_j - t_{j-1})} (E+B)^{-1} \right] e^{A(t_k-s)} C \right] \\
&\quad \times \left[x_\alpha^* \prod_{j=1}^{k} \left[e^{-A(t_j - t_{j-1})} (E+B)^{-1} \right] e^{A(t_k-s)} C \right]^* ds. \qquad (23)
\end{aligned}
$$

According to (23), we obtain

$$
\begin{cases}
x_\alpha^* e^{A(t_0-t)} C = 0, \ \ t \in [t_0, t_1], \\
x_\alpha^* \prod_{j=1}^{i-1} \left[e^{-A(t_j - t_{j-1})} (E+B)^{-1} \right] e^{A(t_{i-1}-t)} C = 0, \ \ t \in (t_{i-1}, t_i], \ \ i = 2, \ldots, k, \\
x_\alpha^* \prod_{j=1}^{k} \left[e^{-A(t_j - t_{j-1})} (E+B)^{-1} \right] e^{A(t_k-t)} C = 0, \ \ t \in (t_k, t_l].
\end{cases} \qquad (24)
$$

Because (1) is state controllable on $[t_0, t_l]$, then there is a piece continuous control function $u(t)$ for initial state $x_0 = x_\alpha$ such that

$$
\begin{aligned}
0 &= x(t_l, x_\alpha) \\
&= e^{A(t_l - t_k)} \prod_{j=k}^{1} \left[(E+B) e^{A(t_j - t_{j-1})} \right] x_\alpha \\
&\quad + e^{A(t_l - t_k)} \sum_{i=1}^{k} \prod_{j=k}^{i} \left[(E+B) e^{A(t_j - t_{j-1})} \right] \int_{t_{i-1}}^{t_i} e^{A(t_{i-1}-s)} C u(s) ds \\
&\quad + \int_{t_k}^{t_l} e^{A(t_l - s)} C u(s) ds \\
&= e^{A(t_l - t_k)} \prod_{j=k}^{1} \left[(E+B) e^{A(t_j - t_{j-1})} \right] x_\alpha \\
&\quad + e^{A(t_l - t_k)} \prod_{j=k}^{1} \left[(E+B) e^{A(t_j - t_{j-1})} \right] \int_{t_0}^{t_1} e^{A(t_0-s)} C u(s) ds
\end{aligned}
$$

$$
\begin{aligned}
&+e^{A(t_l-t_k)}\sum_{i=2}^{k}\prod_{j=k}^{i}\left[(E+B)e^{A(t_j-t_{j-1})}\right]\int_{t_{i-1}}^{t_i}e^{A(t_{i-1}-s)}Cu(s)ds\\
&+\int_{t_k}^{t_l}e^{A(t_l-s)}Cu(s)ds. \qquad (25)
\end{aligned}
$$

From (25), we derive

$$
\begin{aligned}
x_\alpha =& -\int_{t_0}^{t_1}e^{A(t_0-s)}Cu(s)ds\\
&-\sum_{i=2}^{k}\prod_{j=1}^{i-1}\left[e^{-A(t_j-t_{j-1})}(E+B)^{-1}\right]\int_{t_{i-1}}^{t_i}e^{A(t_{i-1}-s)}Cu(s)ds\\
&-\prod_{j=1}^{k}\left[e^{-A(t_j-t_{j-1})}(E+B)^{-1}\right]\int_{t_k}^{t_l}e^{A(t_k-s)}Cu(s)ds. \qquad (26)
\end{aligned}
$$

By multiplying x_α^* to both sides of (26) from left, from (24), we obtain

$$
\begin{aligned}
x_\alpha^*x_\alpha =& -\int_{t_0}^{t_1}x_\alpha^*e^{A(t_0-s)}Cu(s)ds\\
&-\sum_{i=2}^{k}x_\alpha^*\prod_{j=1}^{i-1}\left[e^{-A(t_j-t_{j-1})}(E+B)^{-1}\right]\int_{t_{i-1}}^{t_i}e^{A(t_{i-1}-s)}Cu(s)ds\\
&-x_\alpha^*\prod_{j=1}^{k}\left[e^{-A(t_j-t_{j-1})}(E+B)^{-1}\right]\int_{t_k}^{t_l}e^{A(t_k-s)}Cu(s)ds\\
=&\ 0.
\end{aligned}
$$

This contradicts $x_\alpha \neq 0$ and we can acquire that $rank(U_0\ U_1\ \cdots\ U_k) = n$.
The proof is completed. □

Appendix E. Proof of Theorem 3.10

Proof. Suppose that (1) is not state controllable, and then according to Theorem 3.4, $\overline{V}_f(f \in \{0, 1, \ldots, k\})$ are not invertible. Namely, $\exists\ \varphi(x_\alpha^*) \neq 0 \in \mathbb{C}^{2n\times 1}$ s.t.

$$
\begin{aligned}
0 &= \varphi(x_\alpha^*)\overline{V}(t_0,t_1)\varphi(x_\alpha)\\
&= \int_{t_0}^{t_1}\varphi(x_\alpha^*)e^{\zeta(A)(t_0-s)}\zeta(C)\zeta(C^*)e^{\zeta(A^*)(t_0-s)}\varphi(x_\alpha)ds\\
&= \int_{t_0}^{t_1}(\varphi(x_\alpha^*)e^{\zeta(A)(t_0-s)}\zeta(C))(\varphi(x_\alpha^*)e^{\zeta(A)(t_0-s)}\zeta(C))^*ds.
\end{aligned}
$$

Therefore,

$$\varphi(x_\alpha^*)e^{\zeta(A)(t_0-t)}\zeta(C) = 0,$$

differentiating $2n-1$ times with respect to t and letting $t = t_0$, we have

$$
\begin{aligned}
\varphi(x_\alpha^*)\zeta(C) &= 0,\\
\varphi(x_\alpha^*)\zeta(A)\zeta(C) &= 0,\\
\varphi(x_\alpha^*)\zeta(A^2)\zeta(C) &= 0,\\
&\vdots\\
\varphi(x_\alpha^*)\zeta(A^{2n-1})\zeta(C) &= 0.
\end{aligned}
$$

Then, we can easily get

$$\varphi(x_\alpha^*)(\zeta(C), \zeta(A)\zeta(C), \ldots, \zeta(A^{2n-1})\zeta(C)) = \varphi(x_\alpha^*)\overline{Q}_c = 0.$$

Since $\varphi(x_\alpha^*) \neq 0$, we get the $rank\overline{Q}_c < 2n$, which means that (10) does not hold. Thus, (1) is state controllable on $[t_0, t_l](t_l \in (t_k, t_{k+1}])$.
The proof is finished. □

Appendix F. Proof of Theorem 3.11

Proof. Suppose that (1) is not state controllable, and then according to Theorem 3.5, $V_f(f \in \{0, 1, \ldots, k\})$ are not invertible. Namely, $\exists\, x_\alpha \neq 0 \in \mathbb{H}^{n\times 1}$ s.t.

$$\begin{aligned} 0 &= x_\alpha^* V(t_0, t_1) x_\alpha \\ &= \int_{t_0}^{t_1} x_\alpha^* e^{A(t_0-s)} C C^* e^{A^*(t_0-s)} x_\alpha ds \\ &= \int_{t_0}^{t_1} (x_\alpha^* e^{A(t_0-s)} C)(x_\alpha^* e^{A(t_0-s)} C)^* ds. \end{aligned}$$

Therefore,

$$x_\alpha^* e^{A(t_0-t)} C = 0,$$

differentiating $2n-1$ times with respect to t and letting $t = t_0$, we have

$$\begin{aligned} x_\alpha^* C &= 0, \\ x_\alpha^* AC &= 0, \\ x_\alpha^* A^2 C &= 0, \\ &\vdots \\ x_\alpha^* A^{2n-1} C &= 0. \end{aligned}$$

Then, we can easily get

$$x_\alpha^*(C, AC, \ldots, A^{2n-1}C) = x_\alpha^* Q_c = 0.$$

Since $x_\alpha^* \neq 0$, we get the $rankQ_c < n$, which means that (11) does not hold. Thus, (1) is state controllable on $[t_0, t_l](t_l \in (t_k, t_{k+1}])$.

The proof is finished. □

Appendix G. Proof of Theorem 3.12

Proof. Assume that

$$rank(\zeta((E+B)^k)\overline{Q}_c, \zeta((E+B)^{k-1})\overline{Q}_c, \ldots, \overline{Q}_c) < 2n.$$

By multiplying $\zeta((E+B)^{-k})$ from the left, the above equation is equivalent to

$$rank(\overline{Q}_c, \zeta((E+B)^{-1})\overline{Q}_c, \ldots, \zeta((E+B)^{-k})\overline{Q}_c) < 2n,$$

and then $\exists\, \varphi(x_\alpha^*) \neq 0 \in \mathbb{C}^{2n\times 1}$ s.t.

$$\varphi(x_\alpha^*)(\overline{Q}_c, \zeta((E+B)^{-1})\overline{Q}_c, \ldots, \zeta((E+B)^{-k})\overline{Q}_c) = 0.$$

It follows from the definition of $\overline{Q}_c$, we obtain that

$$\varphi(x_\alpha^*)\zeta((E+B)^{1-i})\zeta(A^j)\zeta(C) = 0,\ i = 1, 2, \ldots, k, k+1,\ j = 0, 1, \ldots, 2n-1.$$

From (8), we have

$$\begin{aligned} \varphi(x_\alpha^*)\overline{U}_0 &= \varphi(x_\alpha^*)\overline{U}(t_0, t_1) \\ &= \int_{t_0}^{t_1} \varphi(x_\alpha^*) e^{\zeta(A)(t_0-s)} \zeta(C)\zeta(C^*) e^{\zeta(A^*)(t_0-s)} ds \\ &= \int_{t_0}^{t_1} \sum_{j=0}^{2n-1} \beta_j(t_0-s)\varphi(x_\alpha^*)\zeta(A^j)\zeta(C)\zeta(C^*) e^{\zeta(A^*)(t_0-s)} ds \\ &= 0, \\ \varphi(x_\alpha^*)\overline{U}_{i-1} &= \varphi(x_\alpha^*)\overline{U}(t_{i-1}, t_i) \\ &= \int_{t_{i-1}}^{t_i} \varphi(x_\alpha^*) \prod_{j=1}^{i-1} \left[e^{-\zeta(A)(t_j-t_{j-1})}\zeta((E+B)^{-1})\right] e^{\zeta(A)(t_{i-1}-s)} \zeta(C) \end{aligned}$$

$$\begin{aligned}
&\times \zeta(C^*)e^{\zeta(A^*)(t_{i-1}-s)}\prod_{j=1}^{i-1}\left[\zeta((E+B^*)^{-1})e^{-\zeta(A^*)(t_j-t_{j-1})}\right]ds\\
&= \int_{t_{i-1}}^{t_i}\varphi(x_\alpha^*)\zeta((E+B)^{1-i})e^{\zeta(A)(t_0-s)}\zeta(C)\\
&\times \zeta(C^*)e^{\zeta(A^*)(t_{i-1}-s)}\prod_{j=1}^{i-1}\left[\zeta((E+B^*)^{-1})e^{-\zeta(A^*)(t_j-t_{j-1})}\right]ds\\
&= \int_{t_{i-1}}^{t_i}\sum_{j=0}^{2n-1}\beta_j(t_0-s)\varphi(x_\alpha^*)\zeta((E+B)^{1-i})\zeta(A^j)\zeta(C)\\
&\times \zeta(C^*)e^{\zeta(A^*)(t_{i-1}-s)}\prod_{j=1}^{i-1}\left[\zeta((E+B^*)^{-1})e^{-\zeta(A^*)(t_j-t_{j-1})}\right]ds\\
&= 0,\\
&\quad i = 2,\ldots,k,
\end{aligned}$$

$$\begin{aligned}
\varphi(x_\alpha^*)\overline{U}_k &= \varphi(x_\alpha^*)\overline{U}(t_k,t_l)\\
&= \int_{t_k}^{t_l}\varphi(x_\alpha^*)\prod_{j=1}^{k}\left[e^{-\zeta(A)(t_j-t_{j-1})}\zeta((E+B)^{-1})\right]e^{\zeta(A)(t_k-s)}\zeta(C)\\
&\times \zeta(C^*)e^{\zeta(A^*)(t_k-s)}\prod_{j=1}^{k}\left[\zeta((E+B^*)^{-1})e^{-\zeta(A^*)(t_j-t_{j-1})}\right]ds\\
&= \int_{t_k}^{t_l}\varphi(x_\alpha^*)\zeta((E+B)^{-k})e^{\zeta(A)(t_0-s)}\zeta(C)\\
&\times \zeta(C^*)e^{\zeta(A^*)(t_k-s)}\prod_{j=1}^{k}\left[\zeta((E+B^*)^{-1})e^{-\zeta(A^*)(t_j-t_{j-1})}\right]ds\\
&= \int_{t_k}^{t_l}\sum_{j=0}^{2n-1}\beta_j(t_0-s)\varphi(x_\alpha^*)\zeta((E+B)^{-k})\zeta(A^j)\zeta(C)\\
&\times \zeta(C^*)e^{\zeta(A^*)(t_k-s)}\prod_{j=1}^{k}\left[\zeta((E+B^*)^{-1})e^{-\zeta(A^*)(t_j-t_{j-1})}\right]ds\\
&= 0,
\end{aligned}$$

and obviously $rank(\overline{U}_0,\overline{U}_1,\ldots,\overline{U}_k) < 2n$, which contradicts Theorem 3.6. Therefore, we have $rank\overline{Z} = rank(\zeta((E+B)^k)\overline{Q}_c,\zeta((E+B)^{k-1})\overline{Q}_c,\ldots,\overline{Q}_c) = 2n$.

The proof is finished. □

Appendix H. Proof of Theorem 3.13

Proof. Assume that

$$rank((E+B)^kQ_c,(E+B)^{k-1}Q_c,\ldots,Q_c) < n.$$

By multiplying $(E+B)^{-k}$ from the left, the above equation is equivalent to

$$rank(Q_c,(E+B)^{-1}Q_c,\ldots,(E+B)^{-k}Q_c) < n,$$

and then $\exists\ x_\alpha^* \neq 0 \in \mathbb{H}^{n\times 1}$ s.t.

$$x_\alpha^*(Q_c,(E+B)^{-1}Q_c,\ldots,(E+B)^{-k}Q_c) = 0.$$

It follows from the definition of Q_c, we obtain that

$$x_\alpha^*(E+B)^{1-i}A^jC = 0,\ \ i = 1,2,\ldots,k,k+1,\ \ j = 0,1,\ldots,2n-1.$$

From (8), we have

$$\begin{aligned}
x_\alpha^*U_0 &= x_\alpha^*U(t_0,t_1)\\
&= \int_{t_0}^{t_1}x_\alpha^*e^{A(t_0-s)}CC^*e^{A^*(t_0-s)}ds
\end{aligned}$$

$$
\begin{aligned}
&= \int_{t_0}^{t_1} \sum_{j=0}^{2n-1} \beta_j(t_0-s) x_\alpha^* A^j C C^* e^{A^*(t_0-s)} ds \\
&= 0, \\
x_\alpha^* U_{i-1} &= x_\alpha^* U(t_{i-1}, t_i) \\
&= \int_{t_{i-1}}^{t_i} x_\alpha^* \prod_{j=1}^{i-1} \left[e^{-A(t_j-t_{j-1})}(E+B)^{-1} \right] e^{A(t_{i-1}-s)} C \\
&\quad \times C^* e^{A^*(t_{i-1}-s)} \prod_{j=1}^{i-1} \left[(E+B^*)^{-1} e^{-A^*(t_j-t_{j-1})} \right] ds \\
&= \int_{t_{i-1}}^{t_i} x_\alpha^* (E+B)^{1-i} e^{A(t_0-s)} C \\
&\quad \times C^* e^{A^*(t_{i-1}-s)} \prod_{j=1}^{i-1} \left[(E+B^*)^{-1} e^{-A^*(t_j-t_{j-1})} \right] ds \\
&= \int_{t_{i-1}}^{t_i} \sum_{j=0}^{2n-1} \beta_j(t_0-s) x_\alpha^* (E+B)^{1-i} A^j C \\
&\quad \times C^* e^{A^*(t_{i-1}-s)} \prod_{j=1}^{i-1} \left[(E+B^*)^{-1} e^{-A^*(t_j-t_{j-1})} \right] ds \\
&= 0, \\
&\quad i = 2, \ldots, k, \\
x_\alpha^* U_k &= x_\alpha^* U(t_k, t_l) \\
&= \int_{t_k}^{t_l} x_\alpha^* \prod_{j=1}^{k} \left[e^{-A(t_j-t_{j-1})}(E+B)^{-1} \right] e^{A(t_k-s)} C \\
&\quad \times C^* e^{A^*(t_k-s)} \prod_{j=1}^{k} \left[(E+B^*)^{-1} e^{-A^*(t_j-t_{j-1})} \right] ds \\
&= \int_{t_k}^{t_l} x_\alpha^* (E+B)^{-k} e^{A(t_0-s)} C \\
&\quad \times C^* e^{A^*(t_k-s)} \prod_{j=1}^{k} \left[(E+B^*)^{-1} e^{-A^*(t_j-t_{j-1})} \right] ds \\
&= \int_{t_k}^{t_l} \sum_{j=0}^{2n-1} \beta_j(t_0-s) x_\alpha^* (E+B)^{-k} A^j C \\
&\quad \times C^* e^{A^*(t_k-s)} \prod_{j=1}^{k} \left[(E+B^*)^{-1} e^{-A^*(t_j-t_{j-1})} \right] ds \\
&= 0,
\end{aligned}
$$

and obviously $rank(U_0, U_1, \ldots, U_k) < n$, which contradicts Theorem 3.7. Therefore, we have $rankZ = rank((E+B)^k Q_c, (E+B)^{k-1} Q_c, \ldots, Q_c) = n$.

The proof is finished. □

Appendix I. Proof of Theorem 4.2

Proof. Owing to (1) and (2), we derive

$$
y(t) = \varphi^{-1}\Bigg\{ \zeta(D) \Bigg[e^{\zeta(A)(t-t_k)} \prod_{j=k}^{1} \left[\zeta((E+B)) e^{\zeta(A)(t_j-t_{j-1})} \right] \varphi(x_0)
$$

$$
+ e^{\zeta(A)(t-t_k)} \sum_{i=1}^{k} \prod_{j=k}^{i} \left[\zeta((E+B))e^{\zeta(A)(t_j-t_{j-1})}\right] \int_{t_{i-1}}^{t_i} e^{\zeta(A)(t_{i-1}-s)}\zeta(C)\varphi(u(s))ds
+ \int_{t_k}^{t} e^{\zeta(A)(t-s)}\zeta(C)\varphi(u(s))ds\Bigg]\Bigg\}.
$$

From Definition 4.1, state observability of (1) is equivalent to state observability of zero-input respond $y(t)$ expressed by

$$
y(t) = \begin{cases}
\varphi^{-1}\left\{\zeta(D)e^{\zeta(A)(t-t_0)}\varphi(x_0)\right\}, \\
t \in [t_0, t_1], \\
\varphi^{-1}\left\{\zeta(D)e^{\zeta(A)(t-t_{i-1})} \prod_{j=i-1}^{1} \left[\zeta((E+B))e^{\zeta(A)(t_j-t_{j-1})}\right]\varphi(x_0)\right\}, \ t \in (t_{i-1}, t_i], \\
i = 2, \ldots, k, \\
\varphi^{-1}\left\{\zeta(D)e^{\zeta(A)(t-t_k)} \prod_{j=k}^{1} \left[\zeta((E+B))e^{\zeta(A)(t_j-t_{j-1})}\right]\varphi(x_0)\right\}, \\
t \in (t_k, t_l].
\end{cases} \tag{27}
$$

Multiplying both sides of (27) respectively, by $e^{\zeta(A^*)(t-t_0)}\zeta(D^*)$ and $\prod_{j=i-1}^{1}\left[e^{\zeta(A^*)(t_j-t_{j-1})} \times\zeta((E+B^*))\right]$ $e^{\zeta(A^*)(t-t_{i-1})}\zeta(D^*)(i = 2, \ldots, k+1)$ from the left and integrating with respect to t_0 from t_l, we obtain

$$
\begin{aligned}
&\int_{t_0}^{t_1} e^{\zeta(A^*)(s-t_0)}\zeta(D^*)\varphi(y(s))ds \\
&+ \sum_{i=2}^{k} \int_{t_{i-1}}^{t_i} \prod_{j=i-1}^{1} \left[e^{\zeta(A^*)(t_j-t_{j-1})}\zeta((E+B^*))\right]e^{\zeta(A^*)(s-t_{i-1})}\zeta(D^*)\varphi(y(s))ds \\
&+ \int_{t_k}^{t_l} \prod_{j=k}^{1}\left[e^{\zeta(A^*)(t_j-t_{j-1})}\zeta((E+B^*))\right]e^{\zeta(A^*)(s-t_k)}\zeta(D^*)\varphi(y(s))ds \\
= &\int_{t_0}^{t_1} e^{\zeta(A^*)(s-t_0)}\zeta(D^*)\zeta(D)e^{\zeta(A)(s-t_0)}\varphi(x_0)ds \\
&+ \sum_{i=2}^{k} \int_{t_{i-1}}^{t_i} \prod_{j=i-1}^{1} \left[e^{\zeta(A^*)(t_j-t_{j-1})}\zeta((E+B^*))\right]e^{\zeta(A^*)(s-t_{i-1})}\zeta(D^*) \\
&\times \zeta(D)e^{\zeta(A)(s-t_{i-1})} \prod_{j=i-1}^{1} \left[\zeta((E+B))e^{\zeta(A)(t_j-t_{j-1})}\right]\varphi(x_0)ds \\
&+ \int_{t_k}^{t_l} \prod_{j=k}^{1}\left[e^{\zeta(A^*)(t_j-t_{j-1})}\zeta((E+B^*))\right]e^{\zeta(A^*)(s-t_k)}\zeta(D^*) \\
&\times \zeta(D)e^{\zeta(A)(s-t_k)} \prod_{j=k}^{1}\left[\zeta((E+B))e^{\zeta(A)(t_j-t_{j-1})}\right]\varphi(x_0)ds \\
= &\Bigg[\int_{t_0}^{t_1} e^{\zeta(A^*)(s-t_0)}\zeta(D^*)\zeta(D)e^{\zeta(A)(s-t_0)}ds \\
&+ \sum_{i=2}^{k} \int_{t_{i-1}}^{t_i} \prod_{j=i-1}^{1} \left[e^{\zeta(A^*)(t_j-t_{j-1})}\zeta((E+B^*))\right]e^{\zeta(A^*)(s-t_{i-1})}\zeta(D^*) \\
&\times \zeta(D)e^{\zeta(A)(s-t_{i-1})} \prod_{j=i-1}^{1} \left[\zeta((E+B))e^{\zeta(A)(t_j-t_{j-1})}\right]ds \\
&+ \int_{t_k}^{t_l} \prod_{j=k}^{1}\left[e^{\zeta(A^*)(t_j-t_{j-1})}\zeta((E+B^*))\right]e^{\zeta(A^*)(s-t_k)}\zeta(D^*)
\end{aligned}
$$

$$
\begin{aligned}
&\times \zeta(D)e^{\zeta(A)(s-t_k)}\prod_{j=k}^{1}\left[\zeta((E+B))e^{\zeta(A)(t_j-t_{j-1})}\right]ds\Bigg]\varphi(x_0)\\
&=\left[\sum_{i=1}^{k}\overline{N}(t_{i-1},t_i)+\overline{N}(t_k,t_l)\right]\varphi(x_0)\\
&=\overline{N}(t_0,t_l)\varphi(x_0).
\end{aligned} \tag{28}
$$

It is obvious that the left-hand side of (28) depends on $y(t)$, $t\in[t_0,t_l]$. So if $\overline{N}(t_0,t_l)$ is invertible, then $x(t_0)=x_0$ is uniquely determined by $y(t)$ for $t\in[t_0,t_l]$.

If the quaternion matrix $\overline{N}(t_0,t_l)$ is not invertible, $\exists\ \varphi(x_\alpha)\in\mathbb{C}^{2n\times 1}\neq 0$ s.t. $\varphi(x_\alpha^*)\overline{N}(t_0,t_l)\varphi(x_\alpha)=0$. Since $\overline{N}(t_{i-1},t_i)(i=1,2,\dots,k)$ and $\overline{N}(t_k,t_l)$ are positive semidefinite matrices, we derive

$$
\begin{aligned}
&\varphi(x_\alpha^*)\overline{N}(t_{i-1},t_i)\varphi(x_\alpha)=0,\ \ i=1,2,\dots,k,\\
&\varphi(x_\alpha^*)\overline{N}(t_k,t_l)\varphi(x_\alpha)=0.
\end{aligned} \tag{29}
$$

Choose $\varphi(x_0)=\varphi(x_\alpha)$, from (27) and (29),

$$
\begin{aligned}
&\int_{t_0}^{t_l}y^*(s)y(s)ds\\
&=\sum_{i=1}^{k}\int_{t_{i-1}}^{t_i}y^*(s)y(s)ds+\int_{t_k}^{t_l}y^*(s)y(s)ds\\
&=\varphi^{-1}\Bigg\{\int_{t_0}^{t_1}\varphi(x_\alpha^*)e^{\zeta(A^*)(s-t_0)}\zeta(D^*)\zeta(D)e^{\zeta(A)(s-t_0)}\varphi(x_\alpha)ds\\
&\quad+\sum_{i=2}^{k}\int_{t_{i-1}}^{t_i}\varphi(x_\alpha^*)\prod_{j=i-1}^{1}\left[e^{\zeta(A^*)(t_j-t_{j-1})}\zeta((E+B^*))\right]e^{\zeta(A^*)(s-t_{i-1})}\zeta(D^*)\\
&\quad\times\zeta(D)e^{\zeta(A)(s-t_{i-1})}\prod_{j=i-1}^{1}\left[\zeta((E+B))e^{\zeta(A)(t_j-t_{j-1})}\right]\varphi(x_\alpha)ds\\
&\quad+\int_{t_k}^{t_l}\varphi(x_\alpha^*)\prod_{j=k}^{1}\left[e^{\zeta(A^*)(t_j-t_{j-1})}\zeta((E+B^*))\right]e^{\zeta(A^*)(s-t_k)}\zeta(D^*)\\
&\quad\times\zeta(D)e^{\zeta(A)(s-t_k)}\prod_{j=k}^{1}\left[\zeta((E+B))e^{\zeta(A)(t_j-t_{j-1})}\right]\varphi(x_\alpha)ds\Bigg\}\\
&=\varphi^{-1}\left\{\varphi(x_\alpha^*)\left[\sum_{i=1}^{k}\overline{N}(t_{i-1},t_i)+\overline{N}(t_k,t_l)\right]\varphi(x_\alpha)\right\}\\
&=\varphi^{-1}\left\{\varphi(x_\alpha^*)\overline{N}(t_0,t_l)\varphi(x_\alpha)\right\}\\
&=0,
\end{aligned}
$$

which implies that $\int_{t_0}^{t_l}\|y(s)\|^2_{\mathbb{H}^{p\times 1}}ds=0$. Thus by (27),

$$
0=y(t)=\begin{cases}
\varphi^{-1}\left\{\zeta(D)e^{\zeta(A)(t-t_0)}\varphi(x_0)\right\},\\
t\in[t_0,t_1],\\
\varphi^{-1}\left\{\zeta(D)e^{\zeta(A)(t-t_{i-1})}\prod_{j=i-1}^{1}\left[\zeta((E+B))e^{\zeta(A)(t_j-t_{j-1})}\right]\varphi(x_0)\right\},\\
t\in(t_{i-1},t_i],\ \ i=2,\dots,k,\\
\varphi^{-1}\left\{\zeta(D)e^{\zeta(A)(t-t_k)}\prod_{j=k}^{1}\left[\zeta((E+B))e^{\zeta(A)(t_j-t_{j-1})}\right]\varphi(x_0)\right\},\\
t\in(t_k,t_l].
\end{cases} \tag{30}
$$

From Definition 4.1 and (30), (1) is not state observable on $[t_0,t_l](t_l\in(t_k,t_{k+1}])$.

The proof is completed. □

Appendix J. Proof of Theorem 4.3

Proof. Owing to (1) and (3), we derive

$$
\begin{aligned}
y(t) = D\Bigg[& e^{A(t-t_k)} \prod_{j=k}^{1} \big[(E+B)e^{A(t_j-t_{j-1})}\big] x_0 \\
& + e^{A(t-t_k)} \sum_{i=1}^{k} \prod_{j=k}^{i} \big[(E+B)e^{A(t_j-t_{j-1})}\big] \int_{t_{i-1}}^{t_i} e^{A(t_{i-1}-s)} Cu(s)ds \\
& + \int_{t_k}^{t} e^{A(t-s)} Cu(s)ds \Bigg].
\end{aligned}
$$

From Definition 4.1, state observability of (1) is equivalent to state observability of zero-input respond $y(t)$ expressed by

$$
y(t) = \begin{cases} De^{A(t-t_0)}x_0, \ \ t \in [t_0, t_1], \\ De^{A(t-t_{i-1})} \displaystyle\prod_{j=i-1}^{1} \big[(E+B)e^{A(t_j-t_{j-1})}\big] x_0, \ \ t \in (t_{i-1}, t_i], \ \ i = 2, \dots, k, \\ De^{A(t-t_k)} \displaystyle\prod_{j=k}^{1} \big[(E+B)e^{A(t_j-t_{j-1})}\big] x_0, \ \ t \in (t_k, t_l]. \end{cases} \tag{31}
$$

Multiplying both sides of (31) respectively, by $e^{A^*(t-t_0)}D^*$ and $\prod_{j=i-1}^{1}\big[e^{A^*(t_j-t_{j-1})}(E+B^*)\big] \times e^{A^*(t-t_{i-1})}D^*(i = 2, \dots, k+1)$ from the left and integrating with respect to t_0 from t_l, we obtain

$$
\begin{aligned}
& \int_{t_0}^{t_1} e^{A^*(s-t_0)} D^* y(s)ds \\
& + \sum_{i=2}^{k} \int_{t_{i-1}}^{t_i} \prod_{j=i-1}^{1} \big[e^{A^*(t_j-t_{j-1})}(E+B^*)\big] e^{A^*(s-t_{i-1})} D^* y(s)ds \\
& + \int_{t_k}^{t_l} \prod_{j=k}^{1} \big[e^{A^*(t_j-t_{j-1})}(E+B^*)\big] e^{A^*(s-t_k)} D^* y(s)ds \\
= & \int_{t_0}^{t_1} e^{A^*(s-t_0)} D^* D e^{A(s-t_0)} x_0 ds \\
& + \sum_{i=2}^{k} \int_{t_{i-1}}^{t_i} \prod_{j=i-1}^{1} \big[e^{A^*(t_j-t_{j-1})}(E+B^*)\big] e^{A^*(s-t_{i-1})} D^* D e^{A(s-t_{i-1})} \prod_{j=i-1}^{1} \big[(E+B)e^{A(t_j-t_{j-1})}\big] x_0 ds \\
& + \int_{t_k}^{t_l} \prod_{j=k}^{1} \big[e^{A^*(t_j-t_{j-1})}(E+B^*)\big] e^{A^*(s-t_k)} D^* D e^{A(s-t_k)} \prod_{j=k}^{1} \big[(E+B)e^{A(t_j-t_{j-1})}\big] x_0 ds \\
= & \Bigg[\int_{t_0}^{t_1} e^{A^*(s-t_0)} D^* D e^{A(s-t_0)} ds \\
& + \sum_{i=2}^{k} \int_{t_{i-1}}^{t_i} \prod_{j=i-1}^{1} \big[e^{A^*(t_j-t_{j-1})}(E+B^*)\big] e^{A^*(s-t_{i-1})} D^* D e^{A(s-t_{i-1})} \prod_{j=i-1}^{1} \big[(E+B)e^{A(t_j-t_{j-1})}\big] ds \\
& + \int_{t_k}^{t_l} \prod_{j=k}^{1} \big[e^{A^*(t_j-t_{j-1})}(E+B^*)\big] e^{A^*(s-t_k)} D^* D e^{A(s-t_k)} \prod_{j=k}^{1} \big[(E+B)e^{A(t_j-t_{j-1})}\big] ds \Bigg] x_0 \\
= & \Bigg[\sum_{i=1}^{k} N(t_{i-1}, t_i) + N(t_k, t_l) \Bigg] x_0 \\
= & N(t_0, t_l) x_0.
\end{aligned} \tag{32}
$$

It is obvious that the left-hand side of (32) depends on $y(t)$, $t \in [t_0, t_l]$. So if $N(t_0, t_l)$ is invertible, then $x(t_0) = x_0$ is uniquely determined by $y(t)$ for $t \in [t_0, t_l]$.

If the quaternion matrix $N(t_0, t_l)$ is not invertible, $\exists\, x_\alpha \in \mathbb{H}^{n\times 1} \neq 0$ s.t. $x_\alpha^* N(t_0, t_l) x_\alpha = 0$. Since $N(t_{i-1}, t_i)(i = 1, 2, \ldots, k)$ and $N(t_k, t_l)$ are positive semidefinite matrices, we derive

$$\begin{aligned} x_\alpha^* N(t_{i-1}, t_i) x_\alpha &= 0,\ i = 1, 2, \ldots, k, \\ x_\alpha^* N(t_k, t_l) x_\alpha &= 0. \end{aligned} \tag{33}$$

Choose $x_0 = x_\alpha$, from (31) and (33),

$$\begin{aligned} &\int_{t_0}^{t_l} y^*(s)y(s)ds \\ &= \sum_{i=1}^{k} \int_{t_{i-1}}^{t_i} y^*(s)y(s)ds + \int_{t_k}^{t_l} y^*(s)y(s)ds \\ &= \int_{t_0}^{t_1} x_\alpha^* e^{A^*(s-t_0)} D^* D e^{A(s-t_0)} x_\alpha ds \\ &\quad + \sum_{i=2}^{k} \int_{t_{i-1}}^{t_i} x_\alpha^* \prod_{j=i-1}^{1} \left[e^{A^*(t_j-t_{j-1})}(E+B^*)\right] e^{A^*(s-t_{i-1})} D^* D e^{A(s-t_{i-1})} \prod_{j=i-1}^{1} \left[(E+B)e^{A(t_j-t_{j-1})}\right] x_\alpha ds \\ &\quad + \int_{t_k}^{t_l} x_\alpha^* \prod_{j=k}^{1} \left[e^{A^*(t_j-t_{j-1})}(E+B^*)\right] e^{A^*(s-t_k)} D^* D e^{A(s-t_k)} \prod_{j=k}^{1} \left[(E+B)e^{A(t_j-t_{j-1})}\right] x_\alpha ds \\ &= x_\alpha^* \left[\sum_{i=1}^{k} N(t_{i-1}, t_i) + N(t_k, t_l)\right] x_\alpha \\ &= x_\alpha^* N(t_0, t_l) x_\alpha \\ &= 0, \end{aligned}$$

which implies that $\int_{t_0}^{t_l} \|y(s)\|^2_{\mathbb{H}^{p\times 1}} ds = 0$. Thus by (31),

$$0 = y(t) = \begin{cases} D e^{A(t-t_0)} x_0,\ t \in [t_0, t_1], \\ D e^{A(t-t_{i-1})} \prod_{j=i-1}^{1} \left[(E+B)e^{A(t_j-t_{j-1})}\right] x_0,\ t \in (t_{i-1}, t_i],\ i = 2, \ldots, k, \\ D e^{A(t-t_k)} \prod_{j=k}^{1} \left[(E+B)e^{A(t_j-t_{j-1})}\right] x_0,\ t \in (t_k, t_l]. \end{cases} \tag{34}$$

From Definition 4.1 and (34), (1) is not state observable on $[t_0, t_l](t_l \in (t_k, t_{k+1}])$.
The proof is completed. □

Appendix K. Proof of Theorem 4.5

Proof. If (1) is not state observable, then the matrix $\overline{N}(t_0, t_l)$ is not invertible, which means that $\exists\ \varphi(x_\alpha) \neq 0$ s.t. $\varphi(x_\alpha^*)\overline{N}(t_0, t_l)\varphi(x_\alpha) = 0$. Since the matrices $\overline{N}(t_{i-1}, t_i)$ are non-negative definite, we get

$$\begin{aligned} \varphi(x_\alpha^*)\overline{N}(t_0, t_1)\varphi(x_\alpha) &= \int_{t_0}^{t_1} \varphi(x_\alpha^*) e^{\zeta(A^*)(s-t_0)} \zeta(D^*)\zeta(D) e^{\zeta(A)(s-t_0)} \varphi(x_\alpha) ds \\ &= \int_{t_0}^{t_1} (\zeta(D) e^{\zeta(A)(s-t_0)} \varphi(x_\alpha))^* (\zeta(D) e^{\zeta(A)(s-t_0)} \varphi(x_\alpha)) ds \\ &= 0. \end{aligned}$$

This shows that

$$\zeta(D) e^{\zeta(A)(t-t_0)} \varphi(x_\alpha) = 0,\ t \in (t_0, t_l]. \tag{35}$$

Differentiating (35) $2n-1$ times with respect to t and letting $t = t_0$, we have

$$\begin{aligned} \zeta(D)\varphi(x_\alpha) &= 0, \\ \zeta(D)\zeta(A)\varphi(x_\alpha) &= 0, \\ \zeta(D)\zeta(A^2)\varphi(x_\alpha) &= 0, \end{aligned}$$

$$\vdots$$
$$\zeta(D)\zeta(A^{2n-1})\varphi(x_\alpha) = 0.$$

Then, we can easily get

$$\begin{pmatrix} \zeta(D) \\ \zeta(D)\zeta(A) \\ \zeta(D)\zeta(A^2) \\ \vdots \\ \zeta(D)\zeta(A^{2n-1}) \end{pmatrix} \varphi(x_\alpha) = \overline{Q}_o\varphi(x_\alpha) = 0.$$

Since $\varphi(x_\alpha) \neq 0$, we get the $rank\overline{Q}_o < 2n$, which leads to a contradiction with (13).
The proof is finished. □

Appendix L. Proof of Theorem 4.6

Proof. If (1) is not state observable, then the matrix $N(t_0, t_l)$ is not invertible, which means that $\exists\ x_\alpha \neq 0$ s.t. $x_\alpha^* N(t_0, t_l)x_\alpha = 0$. Since the matrices $N(t_{i-1}, t_i)$ are non-negative definite, we get

$$\begin{aligned} x_\alpha^* N(t_0, t_1)x_\alpha &= \int_{t_0}^{t_1} x_\alpha^* e^{A^*(s-t_0)}D^*De^{A(s-t_0)}x_\alpha ds \\ &= \int_{t_0}^{t_1} (De^{A(s-t_0)}x_\alpha)^*(De^{A(s-t_0)}x_\alpha)ds \\ &= 0. \end{aligned}$$

This shows that

$$De^{A(t-t_0)}x_\alpha = 0,\ \ t \in (t_0, t_l]. \tag{36}$$

Differentiating (36) $2n-1$ times with respect to t and letting $t = t_0$, we have

$$\begin{aligned} Dx_\alpha &= 0, \\ DAx_\alpha &= 0, \\ DA^2x_\alpha &= 0, \\ &\vdots \\ DA^{2n-1}x_\alpha &= 0. \end{aligned}$$

Then, we can easily get

$$\begin{pmatrix} D \\ DA \\ DA^2 \\ \vdots \\ DA^{2n-1} \end{pmatrix} x_\alpha = Q_o x_\alpha = 0.$$

Since $x_\alpha \neq 0$, we get the $rankQ_o < n$, which leads to a contradiction with (14).
The proof is finished. □

Appendix M. Proof of Theorem 4.7

Proof. If otherwise, suppose that (1) is state observable on $[t_0, t_l](t_l \in (t_k, t_{k+1}])$, while $rank(\overline{S}) < 2n$, then $\exists\ \varphi(x_\alpha) \neq 0$ s.t.

$$\begin{pmatrix} \overline{Q}_o \\ \overline{Q}_o\zeta((E+B)) \\ \vdots \\ \overline{Q}_o\zeta((E+B)^k) \end{pmatrix} \varphi(x_\alpha) = 0.$$

It follows from the definition of $\overline{Q}_0$, we obtain that

$$\zeta(D)\zeta(A^j)\zeta((E+B)^{i-1})\varphi(x_\alpha)=0,\ \ i=1,2,\ldots,k,k+1,\ \ j=0,1,\ldots,2n-1. \tag{37}$$

By (5), (12) and (37),

$$\begin{aligned}
&\overline{N}(t_0,t_l)\varphi(x_\alpha)\\
=&\left[\sum_{i=1}^{k}\overline{N}(t_{i-1},t_i)+\overline{N}(t_k,t_l)\right]\varphi(x_\alpha)\\
=&\Bigg[\int_{t_0}^{t_1}e^{\zeta(A^*)(s-t_0)}\zeta(D^*)\zeta(D)e^{\zeta(A)(s-t_0)}ds\\
&+\sum_{i=2}^{k}\int_{t_{i-1}}^{t_i}\prod_{j=i-1}^{1}\left[e^{\zeta(A^*)(t_j-t_{j-1})}\zeta((E+B^*))\right]e^{\zeta(A^*)(s-t_{i-1})}\zeta(D^*)\\
&\times\zeta(D)e^{\zeta(A)(s-t_{i-1})}\prod_{j=i-1}^{1}\left[\zeta((E+B))e^{\zeta(A)(t_j-t_{j-1})}\right]ds\\
&+\int_{t_k}^{t_l}\prod_{j=k}^{1}\left[e^{\zeta(A^*)(t_j-t_{j-1})}\zeta((E+B^*))\right]e^{\zeta(A^*)(s-t_k)}\zeta(D^*)\\
&\times\zeta(D)e^{\zeta(A)(s-t_k)}\prod_{j=k}^{1}\left[\zeta((E+B))e^{\zeta(A)(t_j-t_{j-1})}\right]ds\Bigg]\varphi(x_\alpha)\\
=&\int_{t_0}^{t_1}e^{\zeta(A^*)(s-t_0)}\zeta(D^*)\zeta(D)e^{\zeta(A)(s-t_0)}\varphi(x_\alpha)ds\\
&+\sum_{i=2}^{k}\int_{t_{i-1}}^{t_i}\prod_{j=i-1}^{1}\left[e^{\zeta(A^*)(t_j-t_{j-1})}\zeta((E+B^*))\right]e^{\zeta(A^*)(s-t_{i-1})}\zeta(D^*)\\
&\times\zeta(D)e^{\zeta(A)(s-t_0)}\zeta((E+B)^{i-1})\varphi(x_\alpha)ds\\
&+\int_{t_k}^{t_l}\prod_{j=k}^{1}\left[e^{\zeta(A^*)(t_j-t_{j-1})}\zeta((E+B^*))\right]e^{\zeta(A^*)(s-t_k)}\zeta(D^*)\\
&\times\zeta(D)e^{\zeta(A)(s-t_0)}\zeta((E+B)^{k})\varphi(x_\alpha)ds\\
=&\int_{t_0}^{t_1}e^{\zeta(A^*)(s-t_0)}\zeta(D^*)\sum_{j=0}^{2n-1}\beta_j(s-t_0)\zeta(D)\zeta(A^j)\varphi(x_\alpha)ds\\
&+\sum_{i=2}^{k}\int_{t_{i-1}}^{t_i}\prod_{j=i-1}^{1}\left[e^{\zeta(A^*)(t_j-t_{j-1})}\zeta((E+B^*))\right]e^{\zeta(A^*)(s-t_{i-1})}\zeta(D^*)\\
&\times\sum_{j=0}^{2n-1}\beta_j(s-t_0)\zeta(D)\zeta(A^j)\zeta((E+B)^{i-1})\varphi(x_\alpha)ds\\
&+\int_{t_k}^{t_l}\prod_{j=k}^{1}\left[e^{\zeta(A^*)(t_j-t_{j-1})}\zeta((E+B^*))\right]e^{\zeta(A^*)(s-t_k)}\zeta(D^*)\\
&\times\sum_{j=0}^{2n-1}\beta_j(s-t_0)\zeta(D)\zeta(A^j)\zeta((E+B)^{k})\varphi(x_\alpha)ds\\
=&\ 0.
\end{aligned}$$

Since $\varphi(x_\alpha)\neq 0$, the matrix $\overline{N}(t_0,t_l)$ is not invertible, by Theorem 4.2, which leads to that (1) is not state observable. This contradicts the assumption.

The proof is finished. □

Appendix N. Proof of Theorem 4.8

Proof. If otherwise, suppose that (1) is state observable on $[t_0, t_l](t_l \in (t_k, t_{k+1}])$, while $rank(S) < n$, then $\exists\, x_\alpha \neq 0$ s.t.

$$\begin{pmatrix} Q_o \\ Q_o(E+B) \\ \vdots \\ Q_o(E+B)^k \end{pmatrix} x_\alpha = 0.$$

It follows from the definition of Q_o, we obtain that

$$DA^j(E+B)^{i-1}x_\alpha = 0,\ i = 1, 2, \dots, k, k+1,\ j = 0, 1, \dots, 2n-1. \tag{38}$$

By (4), (12) and (38),

$$\begin{aligned}
&N(t_0, t_l)x_\alpha \\
&= \left[\sum_{i=1}^{k} N(t_{i-1}, t_i) + N(t_k, t_l)\right]x_\alpha \\
&= \Bigg[\int_{t_0}^{t_1} e^{A^*(s-t_0)}D^*De^{A(s-t_0)}ds \\
&\quad + \sum_{i=2}^{k}\int_{t_{i-1}}^{t_i}\prod_{j=i-1}^{1}\left[e^{A^*(t_j-t_{j-1})}(E+B^*)\right]e^{A^*(s-t_{i-1})}D^* \\
&\quad \times De^{A(s-t_{i-1})}\prod_{j=i-1}^{1}\left[(E+B)e^{A(t_j-t_{j-1})}\right]ds \\
&\quad + \int_{t_k}^{t_l}\prod_{j=k}^{1}\left[e^{A^*(t_j-t_{j-1})}(E+B^*)\right]e^{A^*(s-t_k)}D^* \\
&\quad \times De^{A(s-t_k)}\prod_{j=k}^{1}\left[(E+B)e^{A(t_j-t_{j-1})}\right]ds\Bigg]x_\alpha \\
&= \int_{t_0}^{t_1} e^{A^*(s-t_0)}D^*De^{A(s-t_0)}x_\alpha ds \\
&\quad + \sum_{i=2}^{k}\int_{t_{i-1}}^{t_i}\prod_{j=i-1}^{1}\left[e^{A^*(t_j-t_{j-1})}(E+B^*)\right]e^{A^*(s-t_{i-1})}D^* \\
&\quad \times De^{A(s-t_0)}(E+B)^{i-1}x_\alpha ds \\
&\quad + \int_{t_k}^{t_l}\prod_{j=k}^{1}\left[e^{A^*(t_j-t_{j-1})}(E+B^*)\right]e^{A^*(s-t_k)}D^* \\
&\quad \times De^{A(s-t_0)}(E+B)^k x_\alpha ds \\
&= \int_{t_0}^{t_1} e^{A^*(s-t_0)}D^*\sum_{j=0}^{2n-1}\beta_j(s-t_0)DA^jx_\alpha ds \\
&\quad + \sum_{i=2}^{k}\int_{t_{i-1}}^{t_i}\prod_{j=i-1}^{1}\left[e^{A^*(t_j-t_{j-1})}(E+B^*)\right]e^{A^*(s-t_{i-1})}D^* \\
&\quad \times \sum_{j=0}^{2n-1}\beta_j(s-t_0)DA^j(E+B)^{i-1}x_\alpha ds \\
&\quad + \int_{t_k}^{t_l}\prod_{j=k}^{1}\left[e^{A^*(t_j-t_{j-1})}(E+B^*)\right]e^{A^*(s-t_k)}D^*
\end{aligned}$$

$$\times \sum_{j=0}^{2n-1} \beta_j(s-t_0)DA^j(E+B)^k x_\alpha ds$$
$$= 0.$$

Since $x_\alpha \neq 0$, the matrix $N(t_0, t_l)$ is not invertible, by Theorem 4.3, which leads to that (1) is not state observable. This contradicts the assumption.

The proof is finished. □

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